



SECJ2203: Software Engineering

System Documentation (SD)

KADA's Cooperative System

Version 1.0

8/12/2024

Faculty of Computing

Prepared by: Group 5

Revision Page

a. Overview

Describe the content of the current version of SD (see below – the note in red).

b. Target Audience

The targeted audiences for this SD are supervisor of the project which is Dr Iqbal Tariq, lecturer of software engineering and stakeholder of the system which is Lembaga Kemajuan Pertanian Kemubu (KADA).

c. Project Team Members

List the team members in a table by stating their roles and the status for each assigned task e.g. by sections for this SD version (complete, partially complete, incomplete). If the assigned tasks are not done and have been assigned to other team members, state accordingly.

Member Name	Role	Task	Status
Syahmi	Leader of the team		
Adam	Co-leader of the team		
Afiq	Manager of the team		
Ziyaad	Member of the team		

d. Version Control History

Version	Primary Author(s)	Description of Version	Date Completed
1.0	Team leader 1 (Full Name)	Completed Chapter X, Section...	dd/mm/yyyy
...			

Table of Contents

1	Introduction		1
	1.1	Purpose	1
	1.2	Scope	?
	1.3	Definitions, Acronyms and Abbreviations	?
	1.4	References	?
	1.5	Overview	?
2	Specific Requirements		?
	2.1	Persona	?
		2.1.1 Persona 1 (User type 1/Actor 1) 2.1.2 Persona 2 (User type 2/Actor 2) 2.1.3 Persona 3 (User type 3/Actor 3) 2.1.n Persona n (User type n/Actor n)	
	2.2	System Features 2.2.1 Use Case Diagram 2.2.2 Domain Model: Class Diagram 2.2.3 State Diagram	
	2.3	Launch Phase	
	2.4	User Story Details	
		2.4.1 US001: User Story <User Story 1> User Story description of US001 Activity Diagram of US001 System Sequence Diagram of US001	

		2.4.2 US002: User Story <User Story 2> User Story description of US002 Activity Diagram of US002 System Sequence Diagram of US002 2.4.3 US002: User Story <User Story 3> User Story description of US003 Activity Diagram of US003 System Sequence Diagram of US003 2.4.n USxxn: User Story <User Story n> User Story description of US00n Activity Diagram of US00n System Sequence Diagram of US00n	
	2.5	Performance and Other Requirements (Non-Functional)	
	2.6	Design Constraints	

1. Introduction

1.1 Purpose

This System Documentation (SD) describes the development, design, and implementation of a digital application portal for Cooperative of KADA. It aims to replace the traditional, manual, paper-based application process with a modernized, user-friendly, and efficient online platform. By leveraging the Systems Development Life Cycle (SDLC) framework, the project seeks to address key challenges such as inefficiency, time consumption, and data security. The documentation provides a detailed account of the project's objectives, methodology, system requirements, and benefits, ensuring transparency and alignment with stakeholders' expectations.

1.2 Scope

The new system for KADA Cooperation is a digital portal designed to simplify and streamline membership and loan applications processes. Applicants can apply for membership online, and once approved, they become members and gain access to loan services. Admins manage and verify applications, while Board Members (ALK) are responsible for making final loan approval decisions. The system improves efficiency, provides secure data storage, and enhances the overall user experience.

Overview of the proposed system

The system will serve four types of users:

1. Applicants:

- Individuals applying for KADA Cooperation membership
- Applicants must submit an online application form and pay the required membership fee
- They receive real-time updates about their application status

2. Members:

- Once approved, applicants become members and can apply for loans anytime
- Members can use the loan calculator to estimate repayment plans and track their loan application statuses.

3. Admins:

- Admins handle all administrative tasks, including verifying membership and loan applications.

- They update applications statuses and forward approved loan applications to the Board Members (ALK) for final decisions.
- Admins also manage user data and ensure smooth operation of the system.

4. **Board Members (ALK):**

- Board Members review loan applications forwarded by Admins.
- They make the final decision on loan approvals or rejection based on eligibility criteria and KADA policies.

Key Features:

- Online membership registration for applicants.
- Loan application and tracking for members.
- Real-time updates on application status via email notifications.
- A loan calculator to help members plan repayment schedules.
- A centralized system for admins and board members to manage and review applications.

Mission Objectives

1. To make KADA's membership and loan applications processes faster and more efficient.
2. To provide a secure and user-friendly platform accessible to applicants, members, admins and board members.
3. To automate routine tasks like status updates and data management, reducing manual errors.
4. To streamline decision-making processes for loan approvals.
5. To support KADA's operational goals with a scalable and efficient system.

Bennefits of the proposed system

Streamlined Processes: Simplifies the processes for membership and loan applications, cutting down the time and effort needed for submissions and evaluations.

User Convenience: Applicants can file forms, monitor application progress, and calculate loan repayments online, eliminating the necessity for face-to-face visits.

Routine Task Automation: Automates the validation of data, updates, and notifications, reducing manual errors and ensuring steady operations.

Enhanced Decision-Making: Offers board members centralized access to reviewed applications and comprehensive reports, increasing the efficiency of loan approvals.

Improve Data Security: Guarantees that sensitive information is securely stored and safeguarded through encryption, adhering to regulatory and legal standards.

Scalable and Future-Ready: Facilitates KADA's expansion with an infrastructure that can handle growing user demand.

1.3 Definitions, Acronyms and Abbreviation

This subsection provides definitions of terms, acronyms, and abbreviations used in the document:

- **KADA:** Lembaga Kemajuan Pertanian Kemubu.
- **ALK:** Ahli Lembaga Koperasi (Board Members).
- **SDLC:** System Development Life Cycle.
- **GDPR:** General Data Protection Regulation.
- **PCI DSS:** Payment Card Industry Data Security Standard.
- **HTTPS:** Hypertext Transfer Protocol Secure

1.4 References

This subsection should:

- a) *Provide a complete list of all documents referenced elsewhere in the SD;*
- b) *Identify each document by title, report number (if applicable), date, and publishing organization;*
- c) *Specify the sources from which the references can be obtained.*

Specify a complete list of references using a standardized reference format.

1.5 Overview

In this System Documentation (SD), we have included specific requirements needed for software development. Specific requirements in this project includes external and internal requirements, system features, performance and other requirements, design constraints and software system attributes. In external interface requirements, we have explained the user interfaces, hardware interfaces, software interfaces and communication interfaces. Every use case has provided use case descriptions and its sequence diagram in system features. Performance and other requirements, design constraints and software system attributes will be elaborated more in this project. All phases are documented with diagrams and descriptions.

2. Specific Requirements

2.1 Persona

The system is designed for multiple user roles, each with unique characteristics, expertise, and needs. Understanding these personas ensures the system is tailored to meet user requirements effectively, enhancing user experience efficiency.

2.1.1 Persona 1 (Applicant)

User Need

- An **applicant** needs a way to submit membership or loan applications online conveniently to avoid physical visits and long processing times.

User Stories

- As an applicant, I want to fill out my application online so that I can avoid traveling to the office.
- As an applicant, I want to check the status of my application online so that I stay informed about its progress.
- As an applicant, I want to receive notifications about application updates so that I know the next steps.
- As an applicant, I want to calculate loan repayments before applying so that I can make informed decisions

2.1.2 Persona 2 (Member)

User Need

- A member needs a way to manage their loan applications, repayment plans, and membership details efficiently.

User Stories

- As a member, I want to apply for loans online so that I can save time and effort.
- As a member, I want to view my loan repayment schedule so that I can plan my finances accordingly.
- As a member, I want to update my personal details so that my information stays accurate.

- As a member, I want to view my membership status and shareholding information so that I stay informed about my contributions.

2.1.3 Persona 3 (Admin)

User Need

- An admin needs a way to efficiently process and manage customer applications and information to ensure a streamlined workflow and reduce manual effort.

User Stories

- As an admin, I want to access and update customer applications easily so that I can ensure all information is accurate and up to date.
- As a admin, I want to review and forward loan applications to the appropriate authorities so that customers receive timely feedback on their loan requests.
- As a admin, I want the system to automatically back up submitted applications so that data loss due to technical issues is minimized.
- As a admin, I want to validate uploaded documents directly within the system so that incomplete applications are flagged for follow-up.
- As a admin, I want to send automated emails for application status updates so that customers are informed without manual intervention.

2.1.4 Persona 4 (ALK / Board Member)

User Need

- A board member needs a way to observe and approve critical decisions, such as loan approvals, to ensure compliance with organizational policies.

User Stories

- As a board member, I want to review high value loan applications so that I ensure compliance with policies.
- As a board member, I want to approve or reject applications forwarded by the admin so that I can oversee critical decisions.
- As a board member, I want to view detailed reports on loans and membership so that I can monitor organizational performance.

- As a board member, I want to communicate decisions to admins promptly so that there are no delays in processing applications.

2.2 System Features

Usecase Diagram

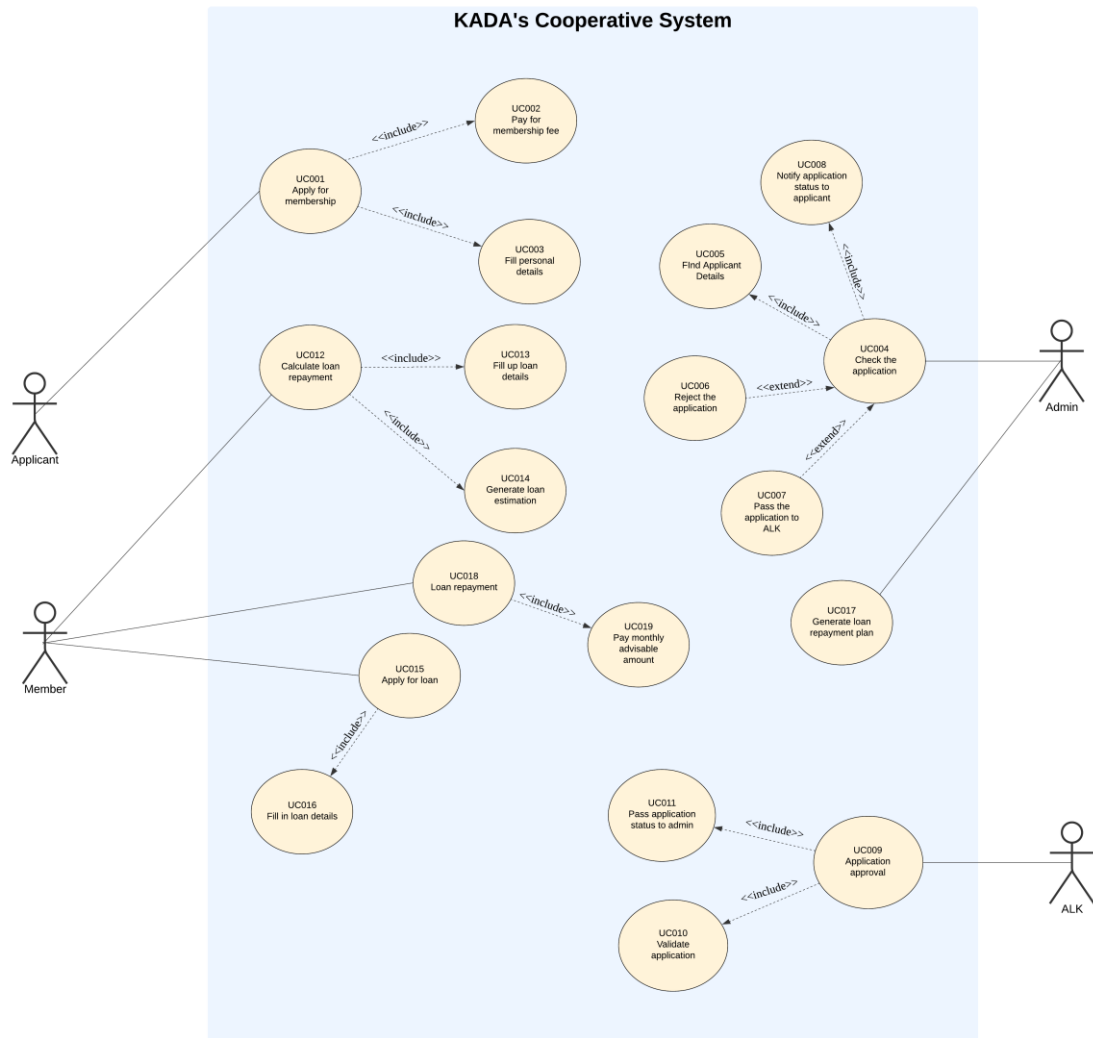


Figure 1.1: Use Case Diagram for KADA's Cooperative System

Table 1.1: Description of Module and Functions for KADA's Cooperative System

Use case	Function	Description
UC001	Apply for membership	Allows an applicant to register as a member by submitting required details
UC002	Pay for membership fee	Facilitates payment of a membership fee to complete the registration
UC003	Fill personal details	Enables applicants to input their personal information during registration
UC004	Check the application	Admin reviews membership or loan application for verification
UC005	Find applicant details	Allows admins to search and retrieve submitted applicant details
UC006	Reject application	Allow admins to reject the member/loan applications of applicants.
UC007	Pass application to ALK	Admin pass the reviewed application from applicant to the ALK
UC008	Notify application status	Admin sends notifications to the applicant regarding their application status.
UC009	Application approval	ALK check and approved the reviewed application pass by the admin.
UC010	Validate application	ALK validates the accuracy and completeness of the application.
UC011	Pass application status	Allows ALK to pass application status to admin after being reviewed
UC012	Calculate loan	Allows member to calculate their total loan repayment using a loan calculator
UC013	Fill loan details	Allows member to fill their desired amount of loan and loan duration to be calculated in the loan calculator
UC014	Generate loan estimation	Generates estimated amount of total loan repayment to member after being calculated
UC015	Apply loan	Allows member to apply for loan

UC016	Fill in loan details	Allows members to fill in loan details.
UC017	Generate loan repayment plan	Allows users to generate loan repayment plans.
UC018	Loan Repayment	Allows lender to view monthly repayment details.
UC019	Pay monthly advisable amount	Allows lender to pay monthly advisable amount.

Domain Model

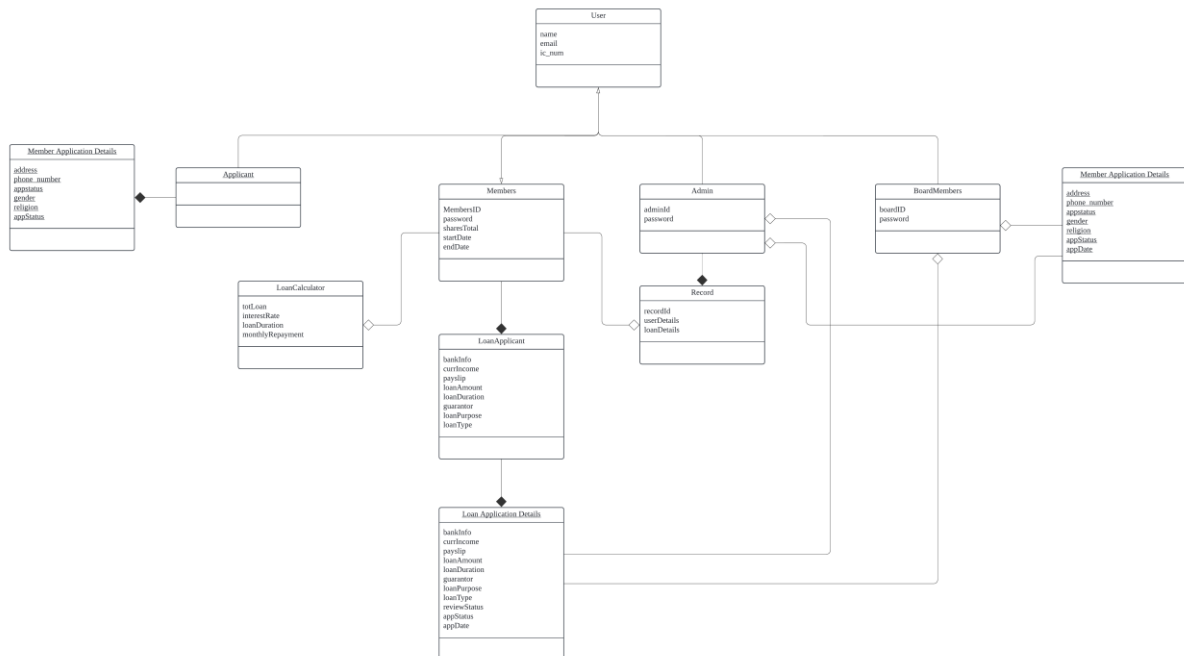


Figure 1.2: Domain Model for KADA's Cooperative System

State Machine Diagram

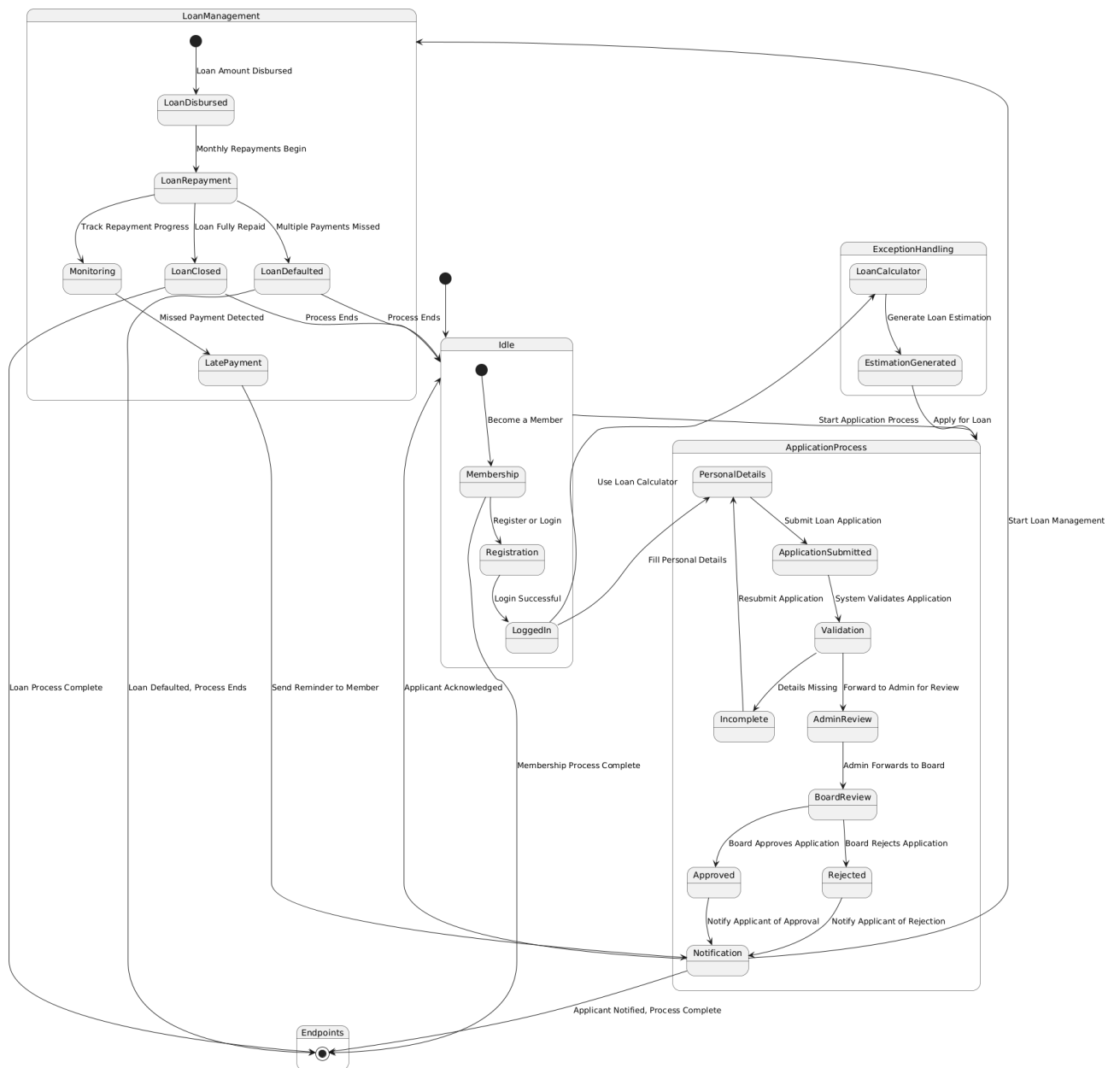


Figure 1.3 State Machine Diagram for KADA's Cooperative system

2.3 User Story Details

This section provides detailed user stories for each use case identified in the system. Each user story outlines the perspective of the intended user, describing their needs, the functionality they require, and the reason for the requirement. These user stories are derived from the use case diagram and are intended to guide the development process by ensuring the system meets user expectations.

2.4.1 US001: User Story <User Story 1>

Table 2.1: User Story Description for Apply for membership

User Story: Apply for Membership
ID: US001
User Story Descriptions As an applicant I want to apply for membership online So that I can access cooperative services efficiently
Flow of events: 1. Applicant navigates to the membership application page. 2. Applicant fills in required personal details and submits the membership form. 3. The system validates the submitted details and confirms the application
Acceptance Criteria: • Precondition: Applicant has access to the system • Postcondition: Membership application is successfully recorded in the system • Other Conditions: Applicant receives confirmation of application submission

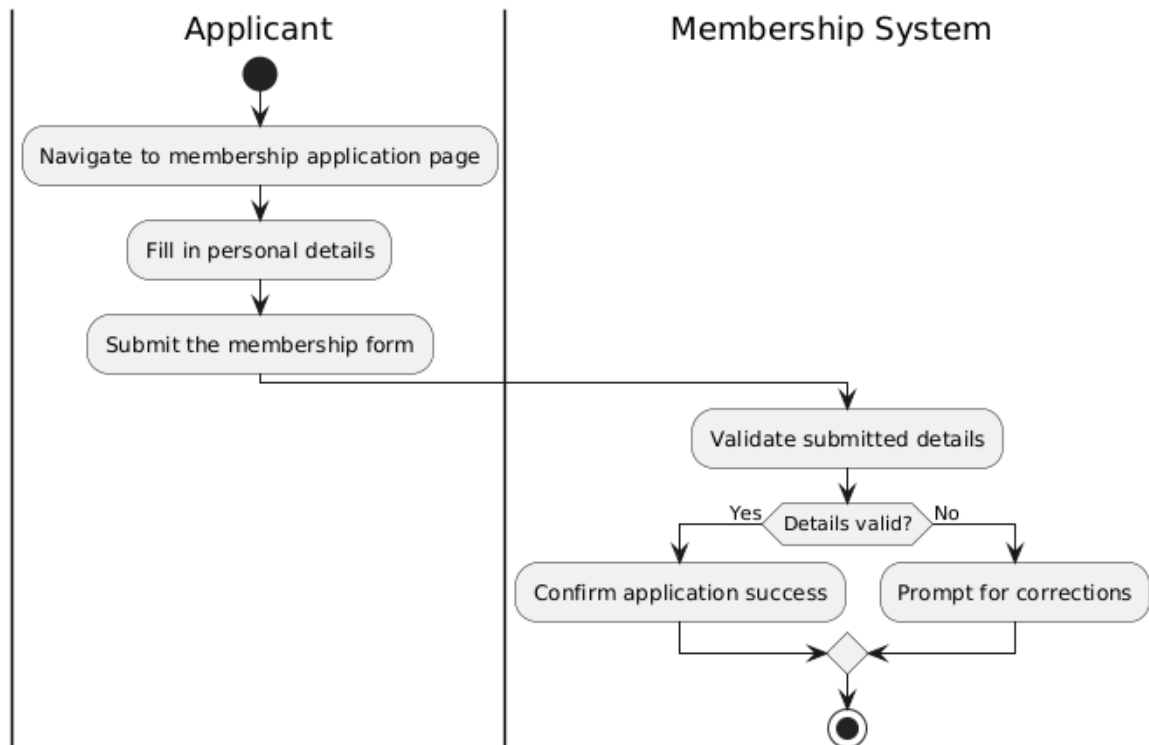


Figure x.x: Activity Diagram of US001

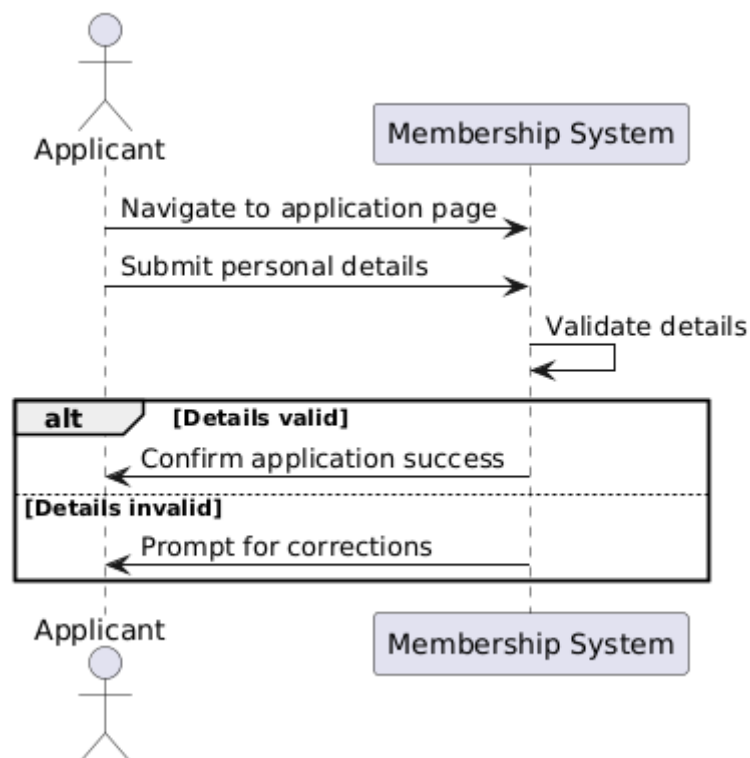


Figure x.x: System Sequence Diagram of US001

2.4.2 US002: User Story <User Story 2>

Table 2.2: User Story Description for Pay for Membership Fee

User Story: Pay for Membership Fee
ID: US002
User Story Descriptions As an applicant I want to pay the membership fee online So that I can complete my registration process conveniently
Flow of events: 1. Applicant proceeds to the payment portal after submitting the application. 2. The system displays available payment options. 3. Applicant upload the proof of the payment, and the system records the transaction.
Acceptance Criteria: • Precondition: Applicant has completed their membership form and payment details • Postcondition: Payment is successfully validated, recorded and membership registration process • Other Conditions: Applicant receives a payment confirmation receipt, and proof of payment is stored securely in the system

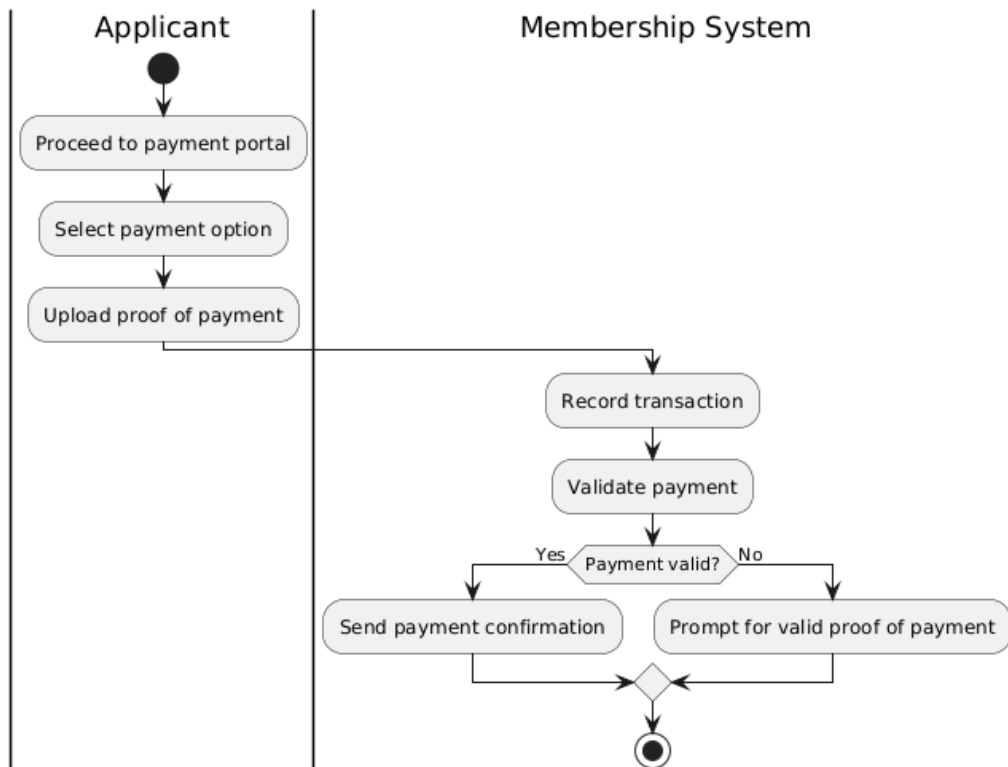


Figure x.x: Activity Diagram of US002

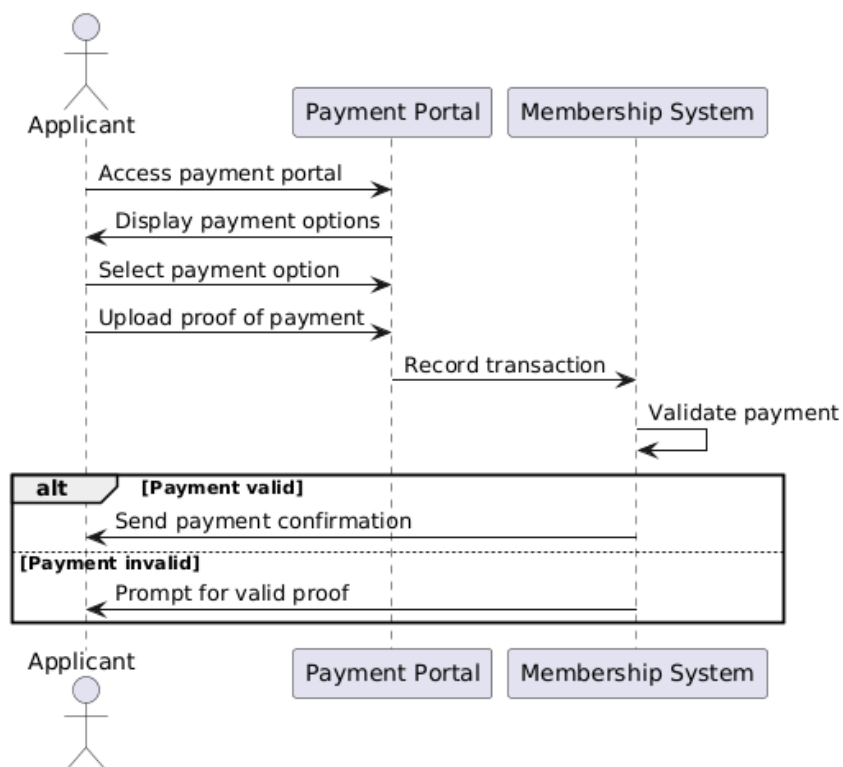


Figure x.x: System Sequence Diagram of US002

2.4.3 US003: User Story <User Story 3>

Table 2.3: User Story Description for Fill Personal Details

User Story: Fill Personal Details
ID: US003
User Story Descriptions As an applicant I want to fill in my personal details accurately So that the system can process my membership or loan application efficiently
Flow of events: 1. Applicant fills in the required personal information fields. 2. The system validates the input fields for completeness and accuracy. 3. Applicant submits the details successfully.
Acceptance Criteria: • Precondition: Applicant is on the personal details page • Postcondition: Personal details are stored in the system database • Other Conditions: System validates all inputs for correctness and completeness

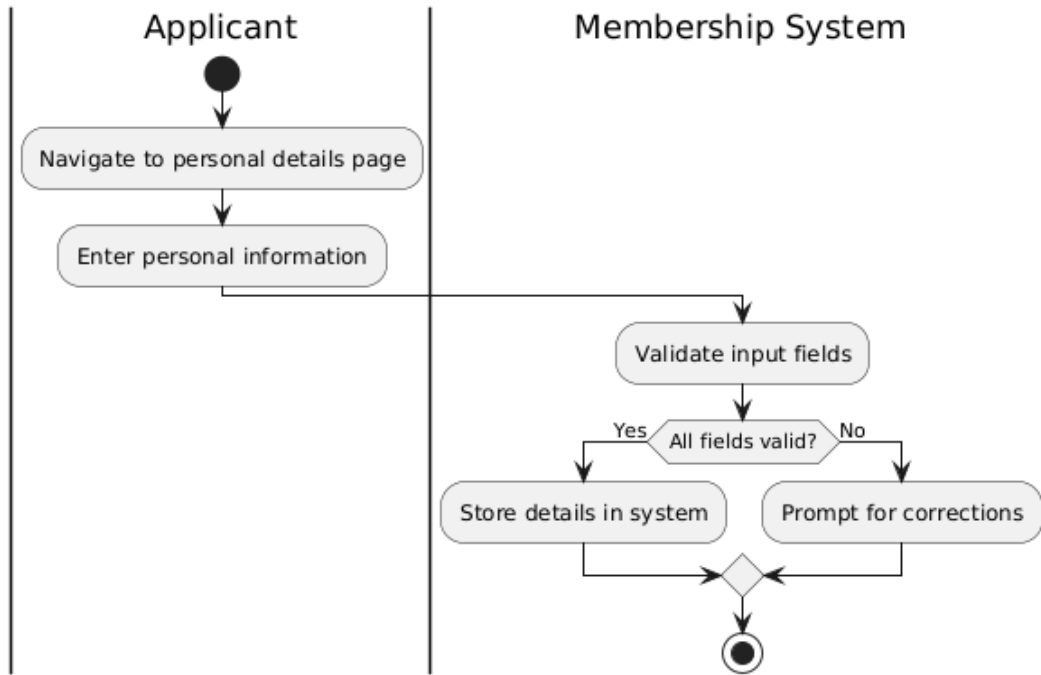


Figure x.x: Activity Diagram of US003

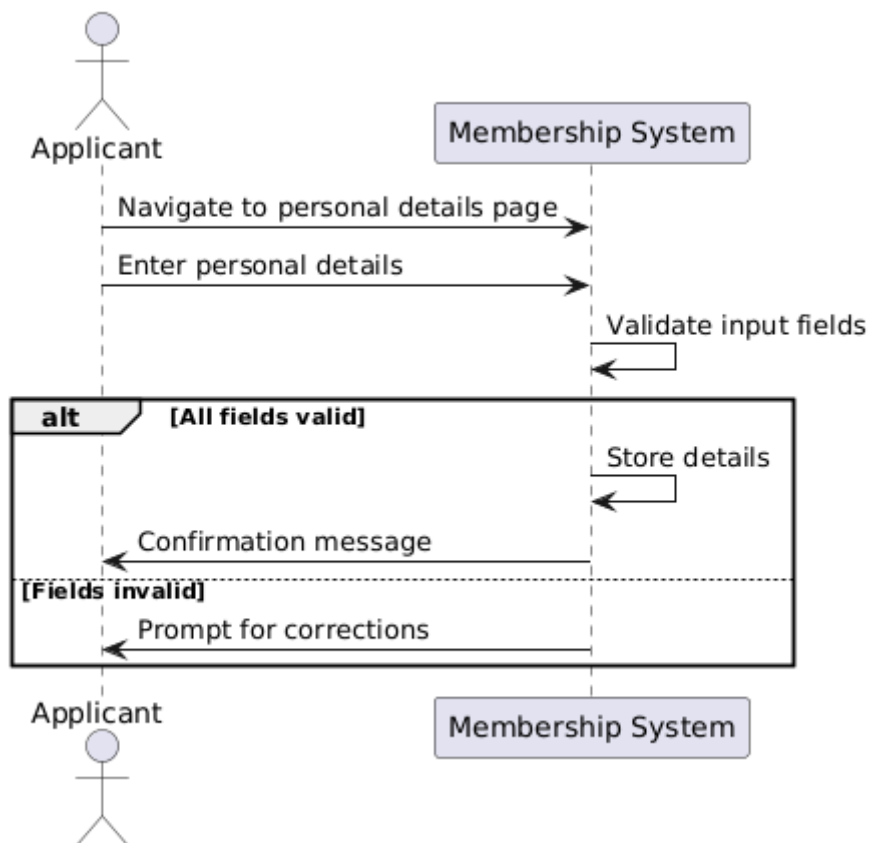


Figure x.x: System Sequence Diagram of US003

2.4.4 US004: User Story <User Story 4>

Table 2.4: User Story Description for Check the Application

User Story: Check the Application
ID: US004
User Story Descriptions As an admin I want to check submitted applications So that I can verify and approve/reject them appropriately
Flow of events: 1. Admin logs into the system and navigates to the applications list. 2. Admin selects an application to review the details. 3. Admin verifies the application and decides whether it should be prioritized based on urgency, completeness of details, and alignment with organizational policies before forwarding to ALK for them to approve or reject the application.
Acceptance Criteria: • Precondition: Admin has access to the applications management section • Postcondition: Application is forwarded to ALK for approval or rejection • Other Conditions: Admin can view prioritization criteria and details about the application

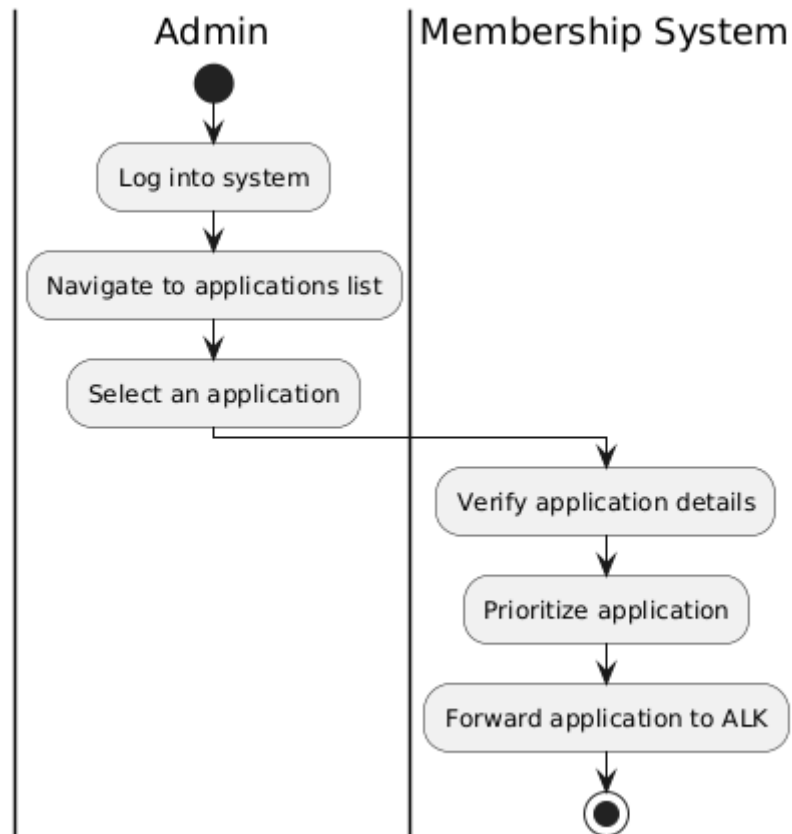


Figure x.x: Activity Diagram of US004

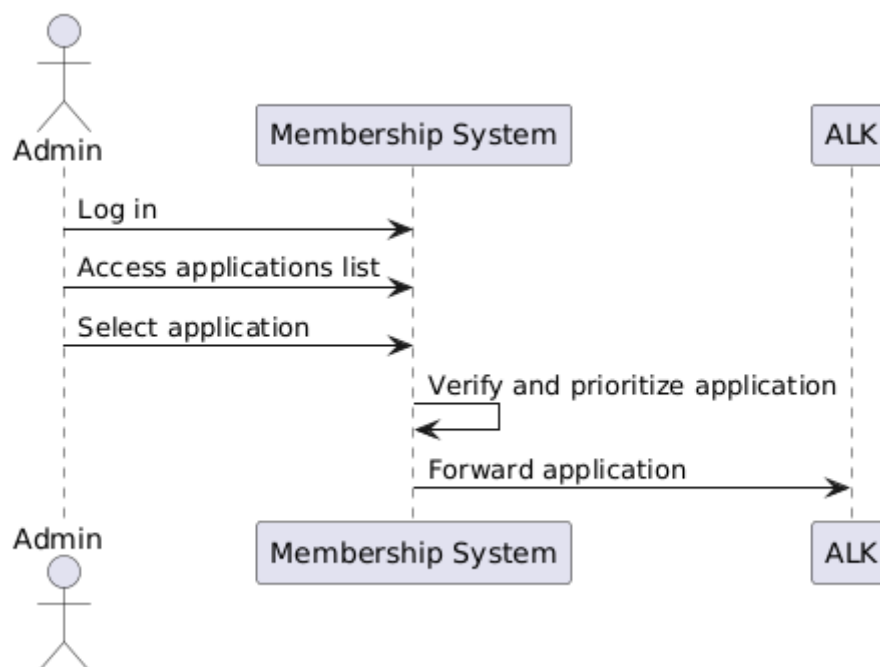


Figure x.x: System Sequence Diagram of US004

2.4.5 US005: User Story <User Story 5>

Table 2.5: User Story Description for Find Applicant Details

User Story: Find Applicant Details
ID: US005
User Story Descriptions As an admin I want to search and retrieve applicant details So that I can manage and verify their applications effectively
Flow of events: 1. Admin logs into the system and navigates to the search functionality. 2. Admin enters relevant search criteria (e.g. name, ID). If no results are found, the system notifies the admin and suggests refining the criteria. If multiple matches are retrieved, the system displays a list for the admin to select the correct applicant. 3. The systems retrieves and displays matching applicant details.
Acceptance Criteria: • Precondition: Admin has system access • Postcondition: Correct applicant details are displayed to the admin • Other Conditions: Search results are accurate and relevant

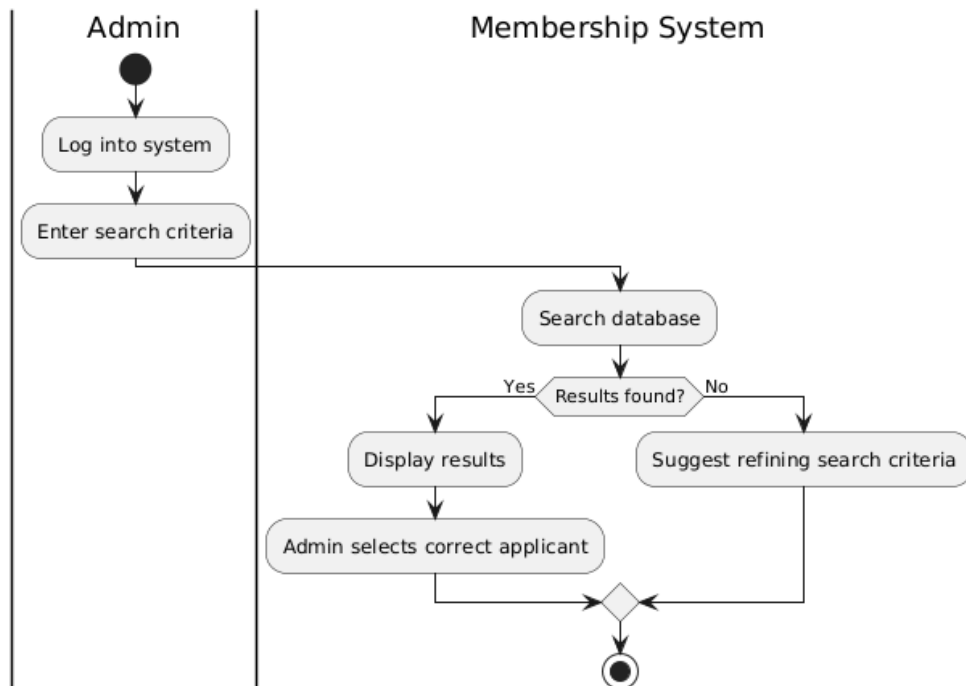


Figure x.x: Activity Diagram of US005

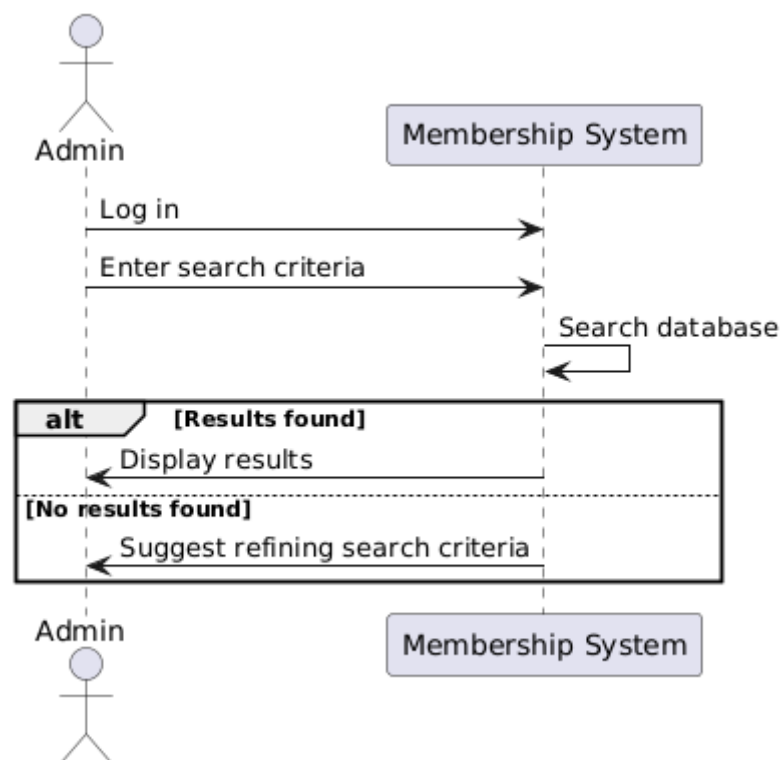


Figure x.x: System Sequence Diagram of US005

2.4.6 US006: User Story <User Story 6>

Table 2.4.6: User Story Description for Rejecting the Application

User Story: Rejecting the application
ID: US006
User Story Descriptions As an admin of KADA cooperative system, if my application of membership is not approved, allow the system to reject the application
Flow of events: 1. Board member receives forwarded application. 2. Board member reviews and decide the status of application 3. Board member reject the application.
Acceptance Criteria: Precondition: The application has been reviewed by the admin. Postcondition: The application status is updated as "Rejected," and the applicant is notified.

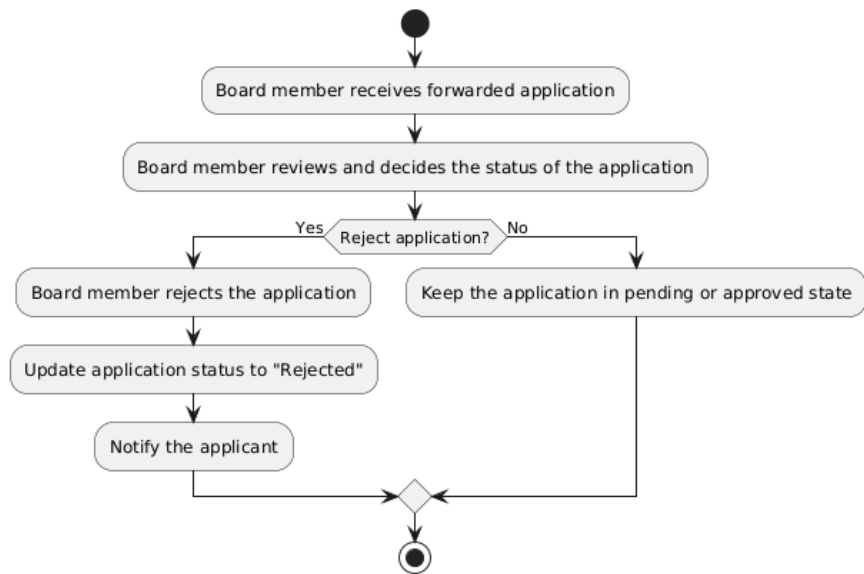


Figure x.x: Activity Diagram of US006

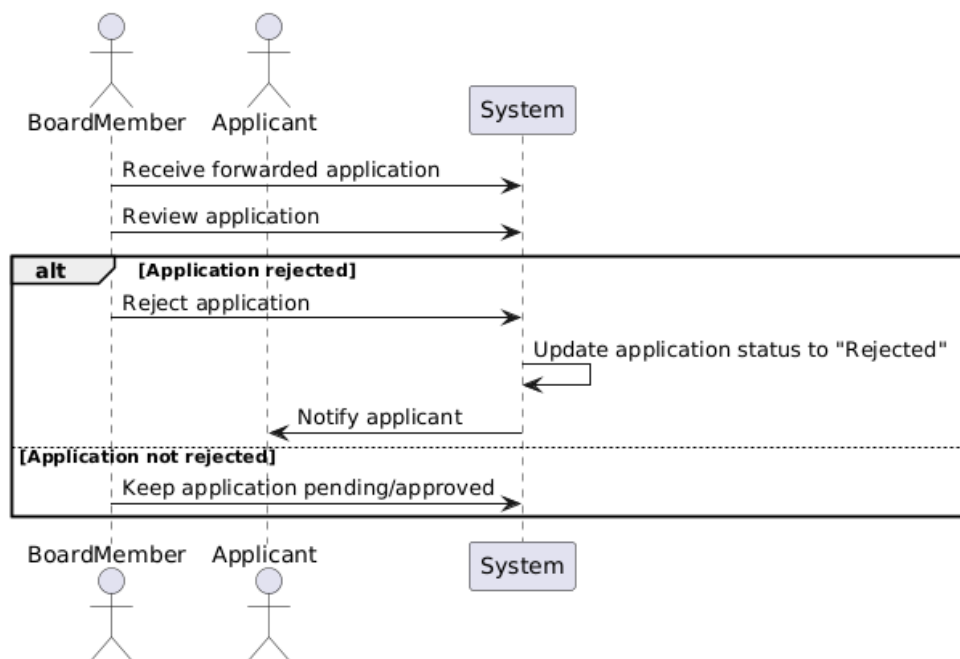


Figure x.x: Sequence Diagram of US006

2.4.7 US007: User Story <User Story 7>

Table 2.4.7: User Story Description for Passing the Application to ALK

User Story: Passing application to ALK
ID: US007
User Story Descriptions As an admin of the cooperative, I want the system to pass the member's application to ALK for further processing.
Flow of events: 1. System receives member application 2. Admin reviews application 3. Admin forward the application to ALK
Acceptance Criteria: Precondition: The application has been received by the system Postcondition: The application is sent to ALK, and the status is updated

Passing Application to ALK

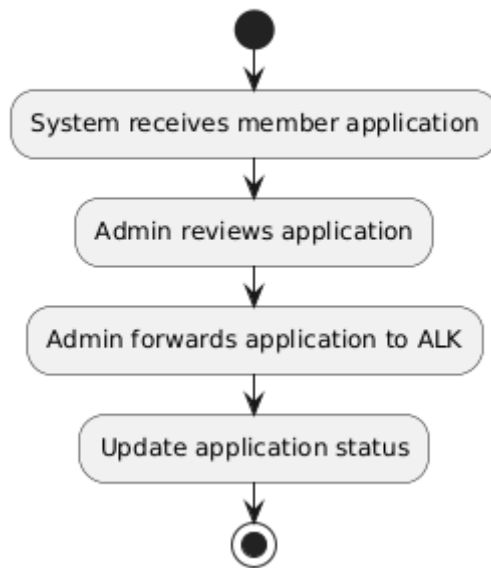


Figure x.x: Activity Diagram of US007

Passing Application to ALK

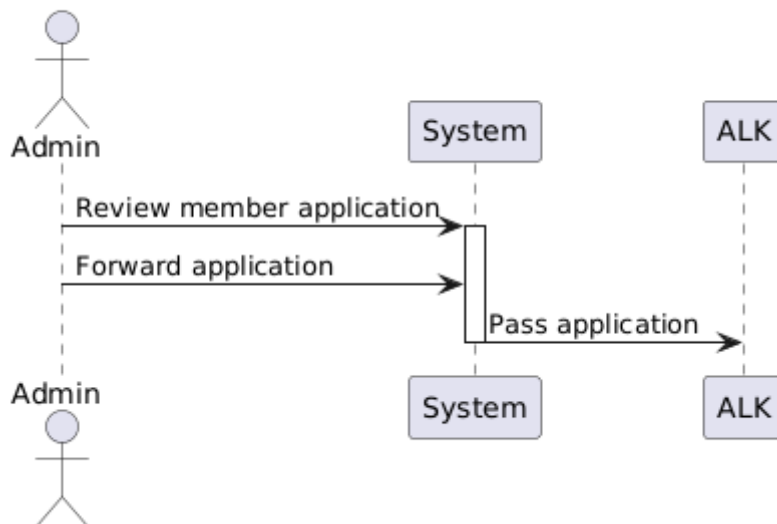


Figure x.x: Sequence Diagram of US007

2.4.8 US008: User Story <User Story 8>

Table 2.4.8: User Story Description for Notifying Application Status to Applicant

User Story: Notifying application status to applicant
ID: US008
User Story Descriptions As an admin of the cooperative, I want the system to notify the applicant of their application status.
Flow of events: 1. Application status is updated from events(UC006, UC007, UC009). 2. System generates a notification message based on the application status received. 3. System send the notification to the applicant via email.
Acceptance Criteria: Precondition: Application status has been successfully updated. Postcondition: Applicant receives the notification.

Notifying Application Status

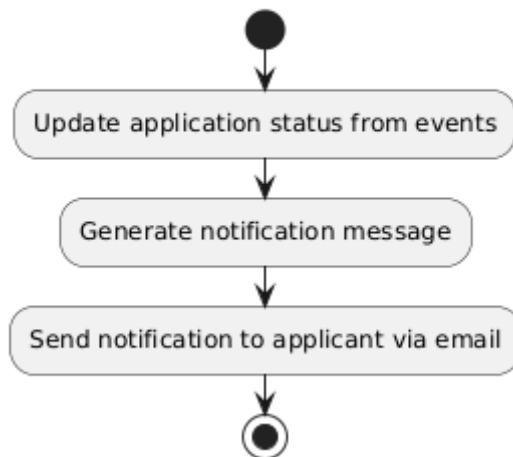


Figure x.x: Activity Diagram of US008

Notifying Application Status

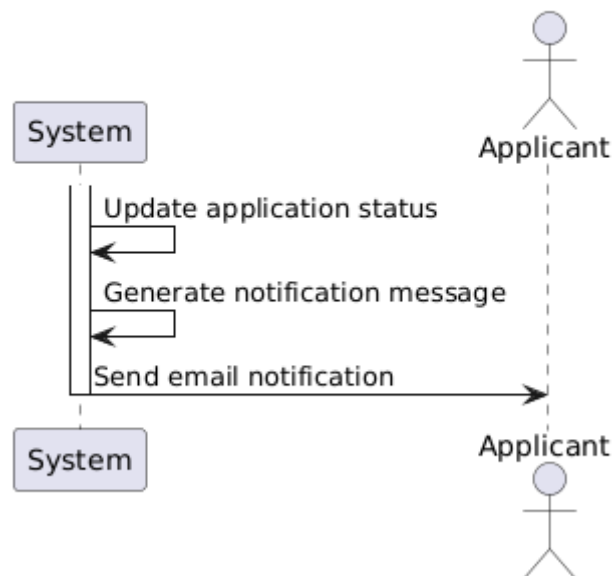


Figure x.x: Sequence Diagram of US008

2.4.9 US009: User Story <User Story 9>

Table 2.4.9: User Story Description for Application Approval

User Story: Application Approval
ID: US009
User Story Descriptions As an ALK of the cooperative, I want the system to approve the member's application.
Flow of events: 1. ALX receives the forwarded application. 2. ALX reviews the application details. 3. ALX approves the application. 4. System updates the application status to "Approved." 5. System triggers UC008 to notify the applicant.
Acceptance Criteria: Precondition: The application has been forwarded to ALX by the admin. Postcondition: The application is marked as "Approved" or "Rejected" and the system triggers a notification to the applicant.

Application Approval Process

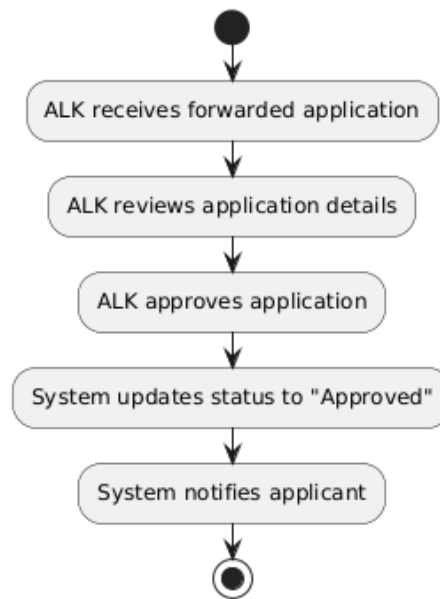


Figure x.x: Activity Diagram of US009

Application Approval Process

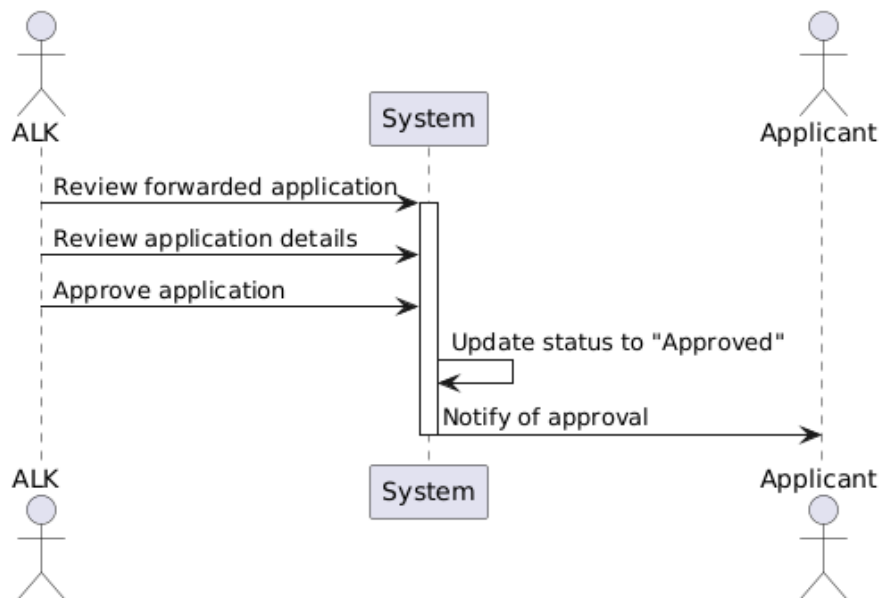


Figure x.x: Sequence Diagram of US009

2.4.10 US010: User Story <User Story 10>

Table 2.4.10: User Story Description for Validating Application

User Story: Validating application
ID: US010
User Story Descriptions As an ALK of the cooperative, I want the system to validate the applicants membership/loan application.
Flow of events: 1. ALX receives the forwarded application. 2. ALX verify the accuracy and the correct criteria of the submitted application. 3. If the application is valid, ALX proceeds to approve it (UC009). 4. If the application is invalid, ALX flags it for further review or rejection.
Acceptance Criteria: Precondition: The application has been forwarded for validation. Postcondition: The application is validated and can proceed to approval (UC009).

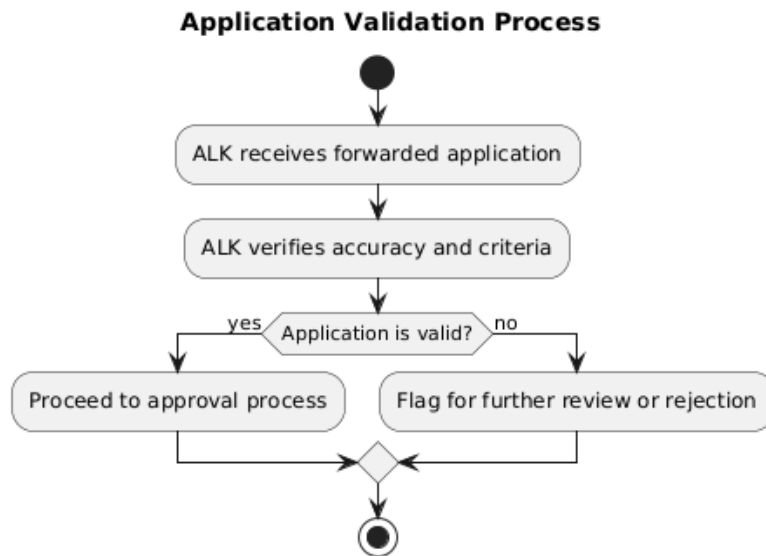


Figure x.x: Activity Diagram of US010

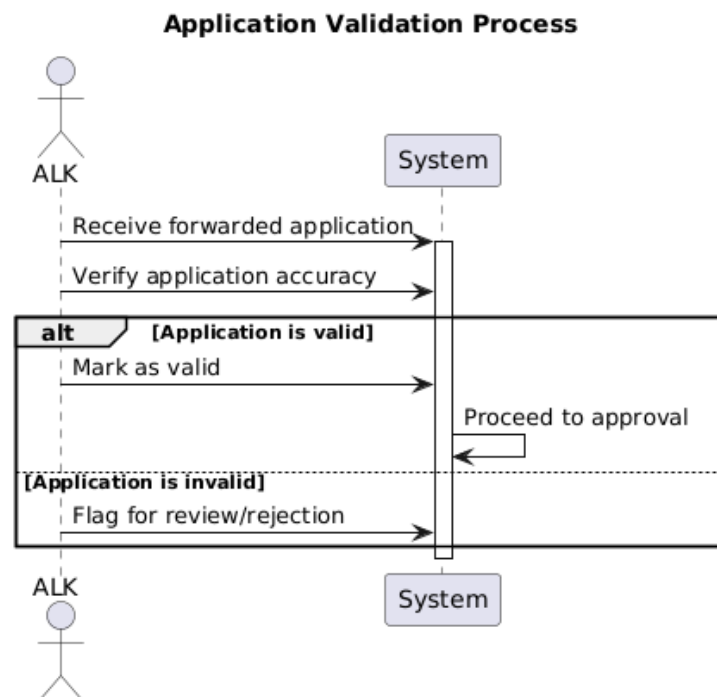


Figure x.x: Sequence Diagram of US010

2.4.11 US011: User Story <User Story 11>

Table 2.1: User Story Description for Pass application status to admin

User story: Pass application status to admin
ID: US011
User Story Description As an ALK (Board Members) of KADA's Cooperative organization, I want to pass the application status to the admin after being reviewed in weekly meetings so that admin can notify the applicants about the result of their application
Flow of events: 1. ALK will select "pass status" buttons. 2. The system will pass the status of application to the admin page.
Acceptance Criteria Precondition: ALK had finalized their decision to either accept or reject the application Postcondition: Admin receives the application status
Exception flow:

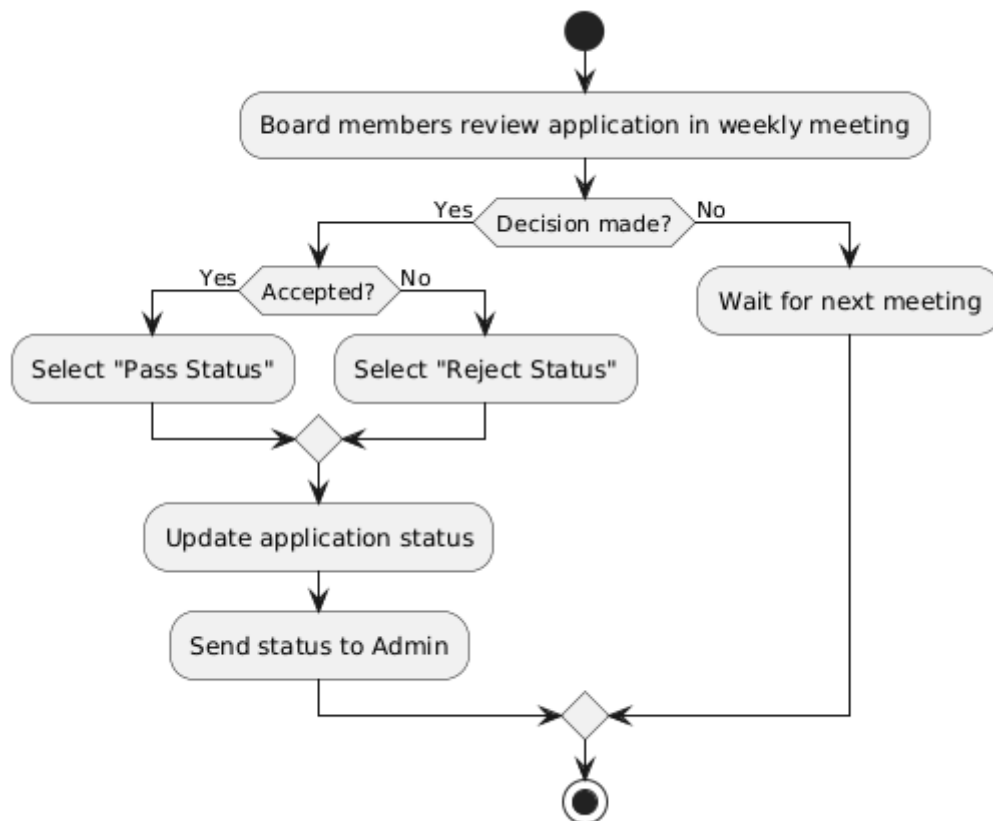


Figure x.x: Activity Diagram of US011

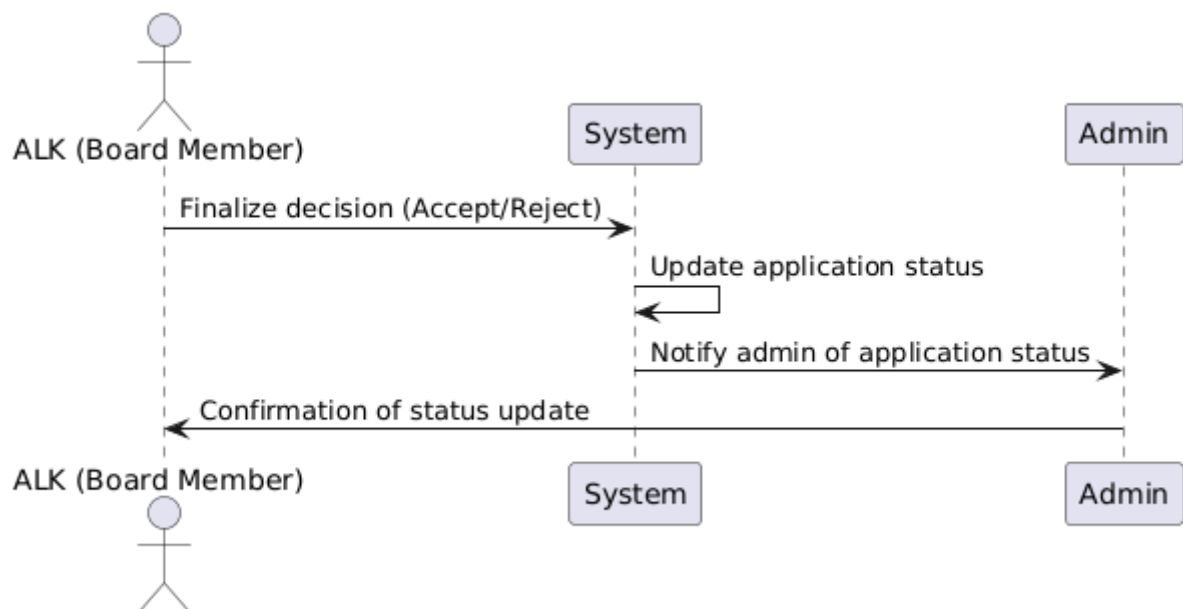


Figure x.x: Sequence Diagram of US011

2.4.12 US012: User Story <User Story 12>

Table 2.1: User Story Description for Calculate loan repayment

User story: Calculate loan repayment
ID: US012
User Story Description As a member, I want to use a loan calculator to calculate total estimate loan repayment so that I can plan my loan amount and duration that is suitable for me.
Flow of events: 1. Member selects “loan calculator”. 2. The system asks the amount of loan desired and duration of the loan. 3. include (Fill up loan details). 4. include (Generate loan estimation).
Alternative flow: 1. At any point the member may move to different page
Acceptance Criteria Precondition: A member logged on to the system. Postcondition: Member receive the amount of estimated loan.
Exception flow:

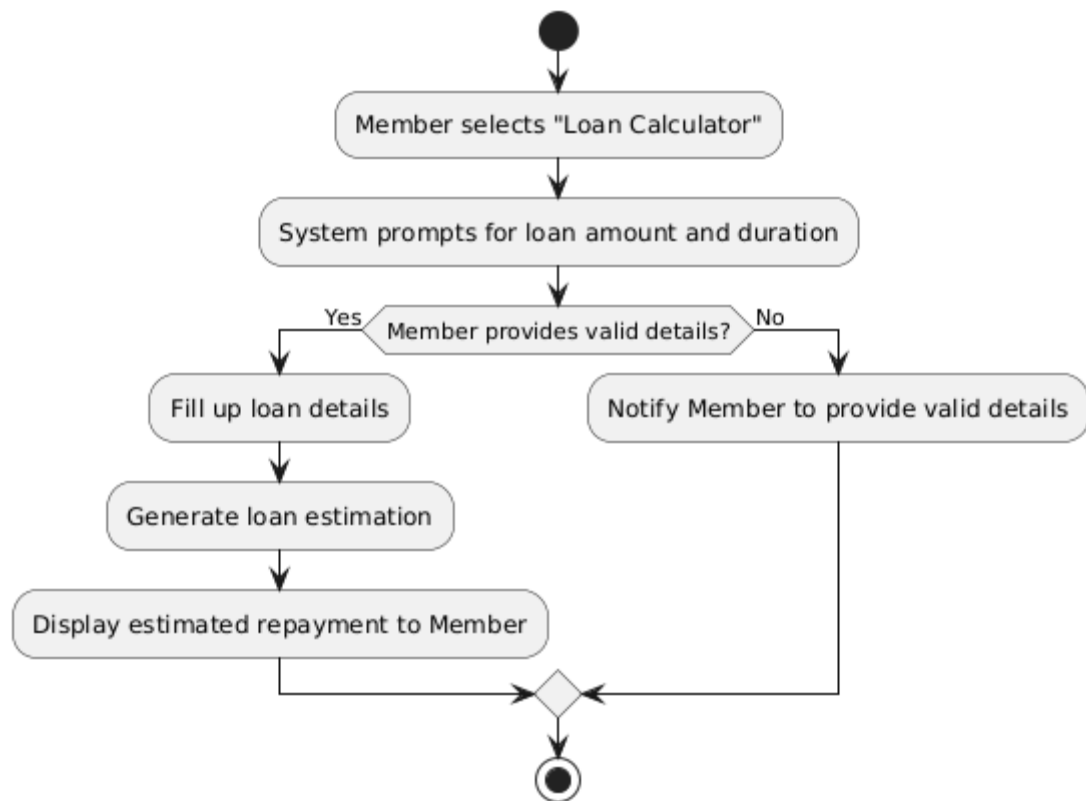


Figure x.x: Activity Diagram of US011

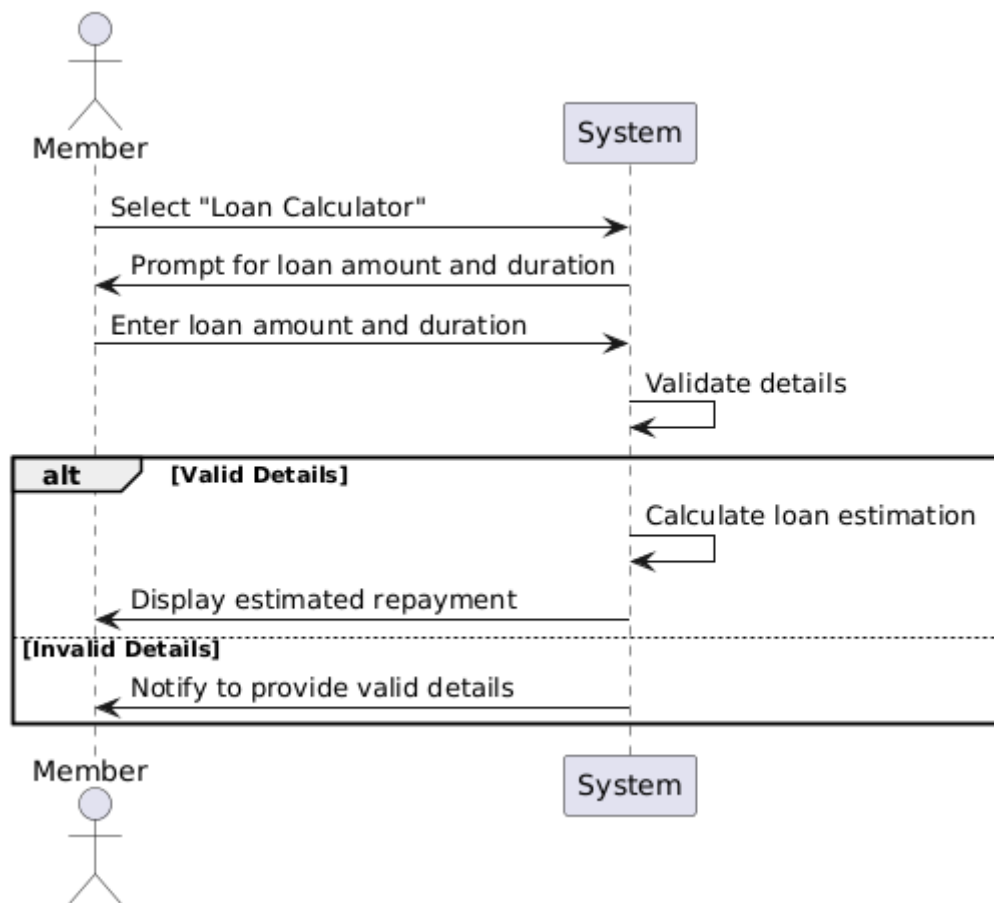


Figure x.x: Sequence Diagram of US012

2.4.13 US013: User Story <User Story 13>

Table 2.1: User Story Description for Fill up loan details

User story: Fill up loan details
ID: US013
User Story Description As a member, I have already selected the loan calculator options. Now, I want to fill up the details about the loan that I want to calculate so that the calculator can give me an estimation for the loan.
Flow of events: 1. The system will ask members to fill in the loan details form. 2. Member enters all the required fields.
Acceptance Criteria Precondition: The member selects “Loan calculator” options.
Exception flow:

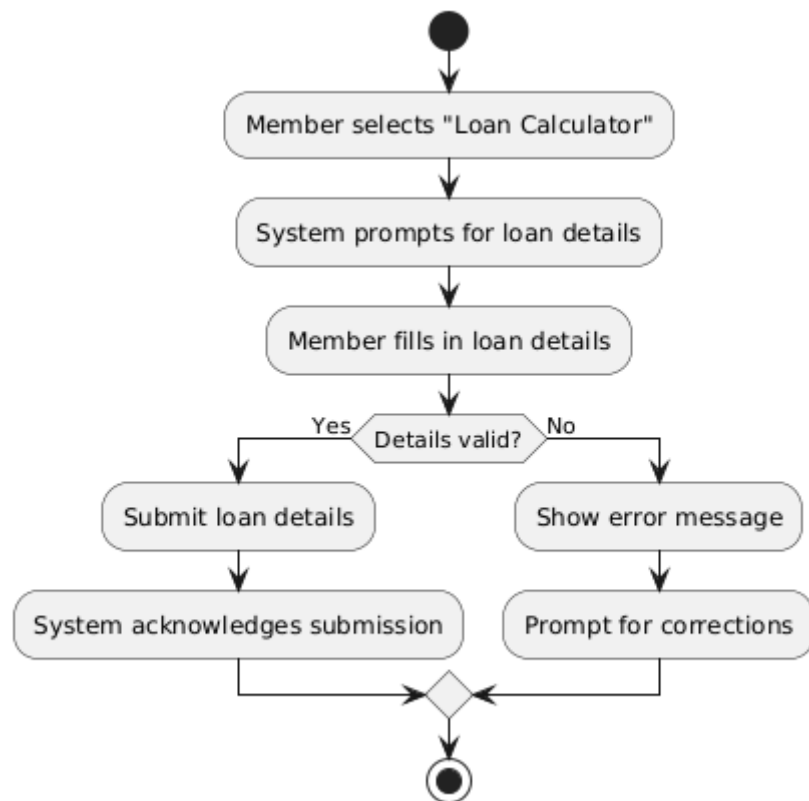


Figure x.x: Activity Diagram of US013

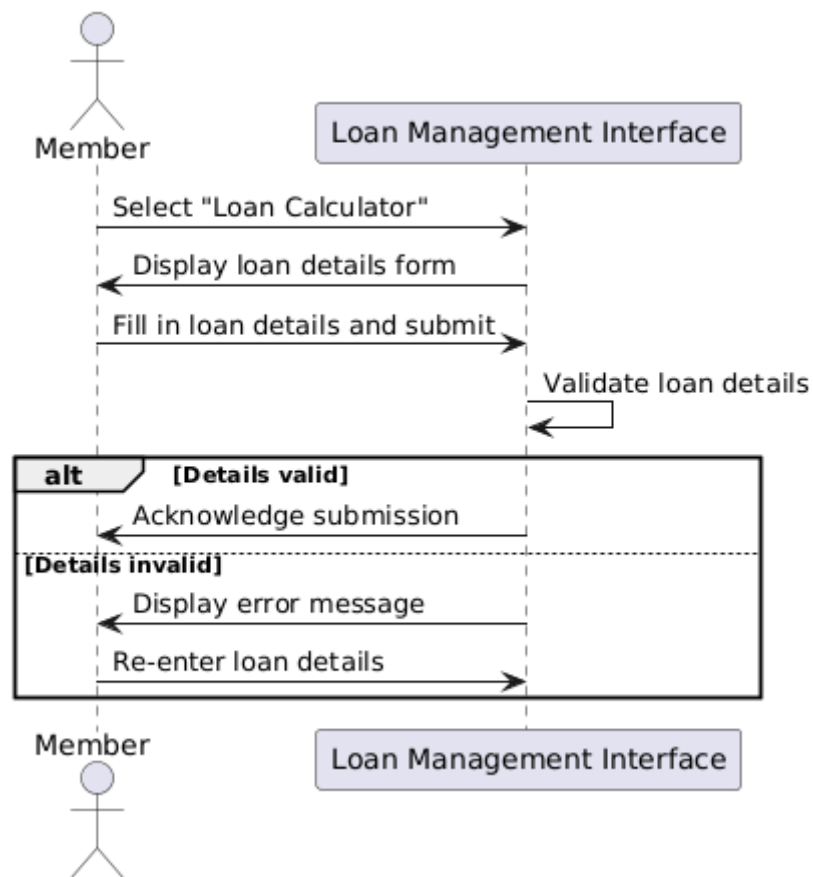


Figure x.x: Sequence Diagram of US013

2.4.14 US014: User Story <User Story 14>

Table 2.1: User Story Description for Generate loan estimation

User story: Generate loan estimation
ID: US014
User Story Description As a member, I already fill up the details of the loan in the loan calculator, and now the calculator will calculate and provide me the estimated amount of loan repayment.
Flow of events: 1. The system will use the member inputs from the form to calculate the total loan repayment using a formula. 2. Then, the system will generate the estimated amount of loan repayment to member.
Acceptance Criteria Precondition: All the required fields already filled by member and member select the “generate” button.
Exception flow:

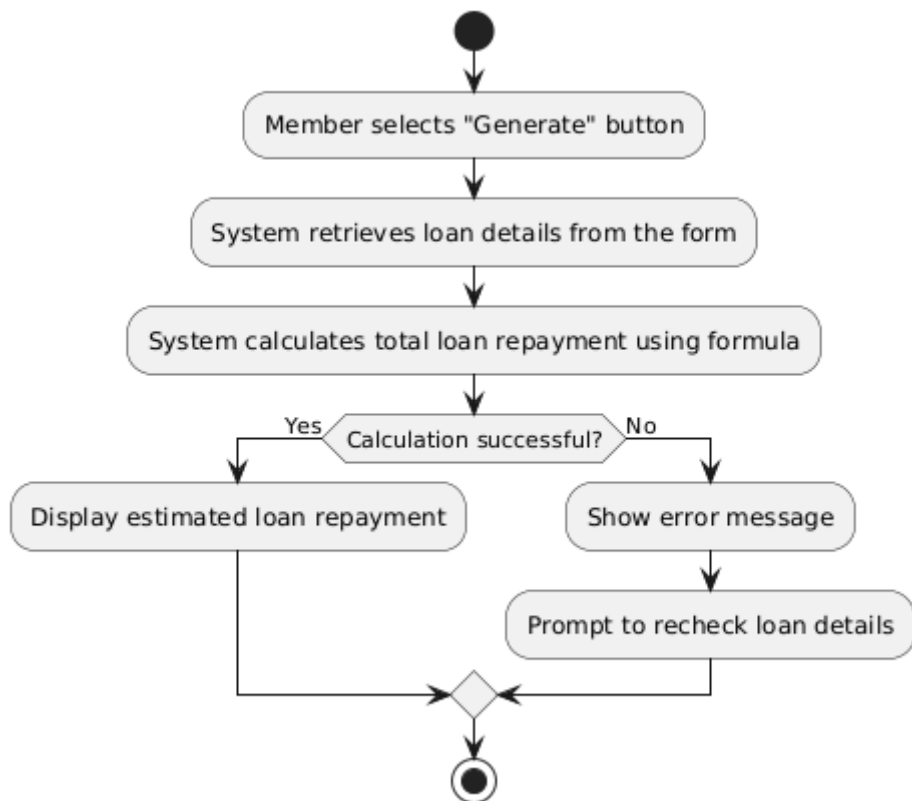


Figure x.x: Activity Diagram of US014

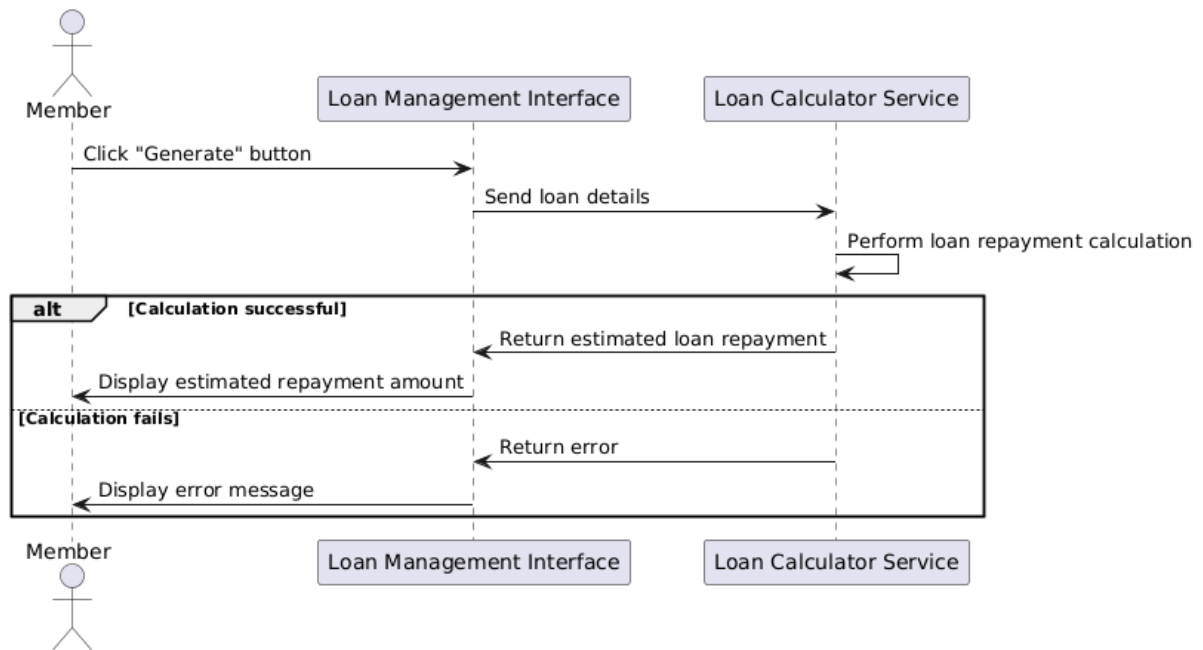


Figure x.x: Sequence Diagram of US014

2.4.15 US015: User Story <User Story 15>

Table 2.1: User Story Description for Apply for loan

User story: Apply for loan
ID: US015
User Story Description As a member, I want to use my benefit to apply for loan from KADA's Cooperative organization, so I can use it for my own personal purposes.
Flow of events: 1. Members select "Apply Loan" options and navigate to the form page. 2. include (Fill in loan details). 3. Submit all the details for admin and ALK to review the application and decides the result.
Alternative flow: 1.Member may skip this option and go to another page.
Acceptance Criteria Precondition: Member must be logged on to the system and select "Apply loan" option. Postcondition: Member need to wait for their application results before creating a new one.
Exception flow:

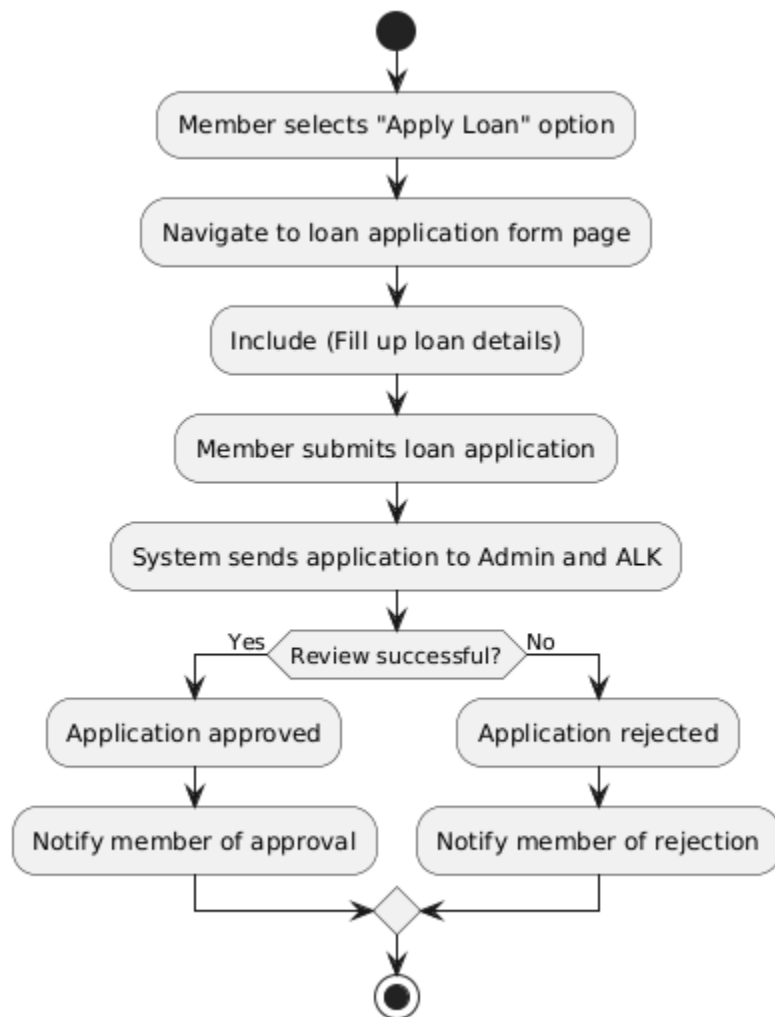


Figure x.x: Activity Diagram of US015

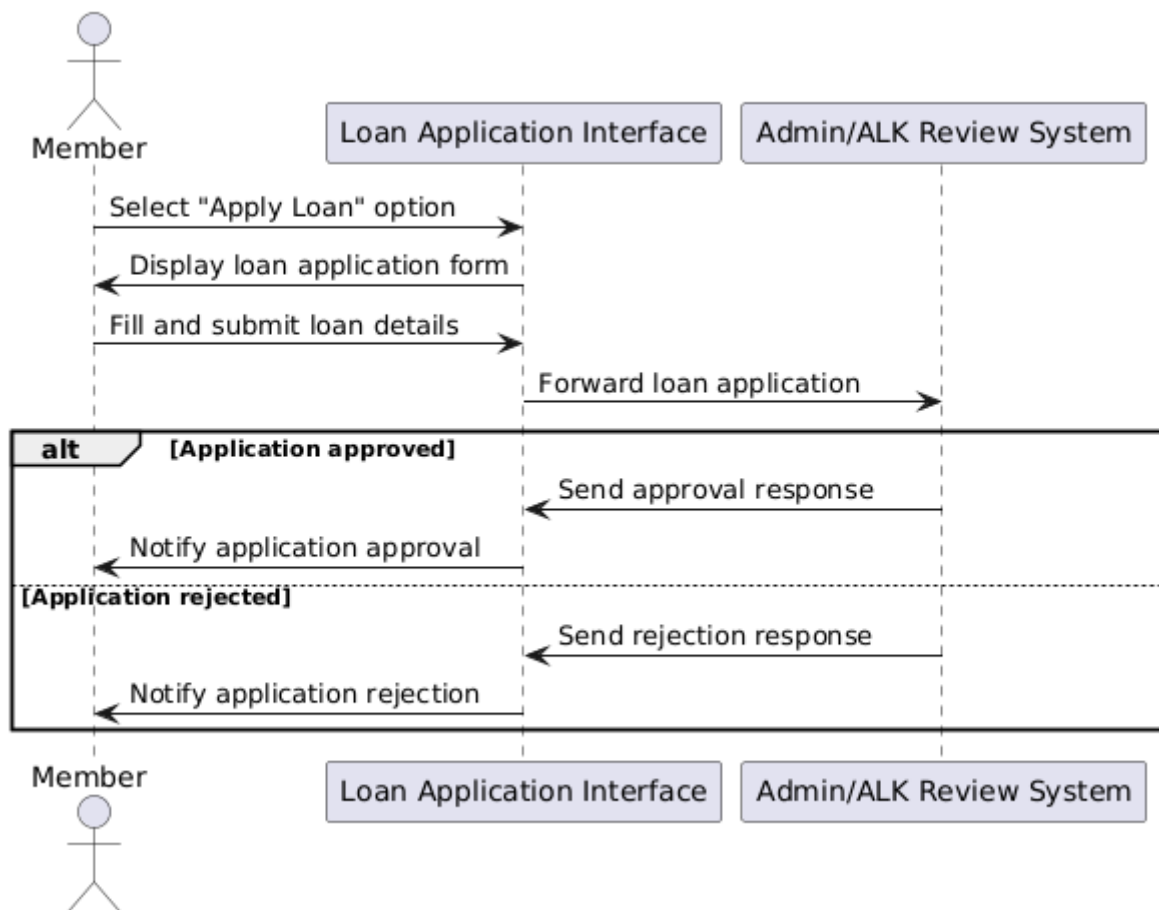


Figure x.x: Sequence Diagram of US015

2.4.16 US016: User Story <User Story 16>

Table 2.1: User Story Description for Fill in loan details

User story: Fill in loan details
ID: US016
User Story Description As a registered member, I want to input my desired loan amount and loan duration, So that the system can calculate my repayment plan.
Flow of events: 1. User logs into the system. 2. User navigates to the "Loan Calculator" section. 3. User selects the "Fill in Loan Details" option. 4. User inputs the desired loan amount. 5. User specifies the loan duration (e.g., number of months or years). 6. System validates the input details and confirms receipt.
Alternative flow: 1. If the loan amount exceeds the allowed limit, the system prompts the user to adjust the input. 2. If the loan duration is invalid, the system displays an error message.
Acceptance Criteria Precondition: The user must be logged in and have no pending loan applications. Postcondition: The system stores the entered loan details for further calculation.
Exception flow: 1. Input errors (e.g., missing or invalid data): The system prompts the user to correct the input. 2. System timeout: The user must re-enter the details.

--

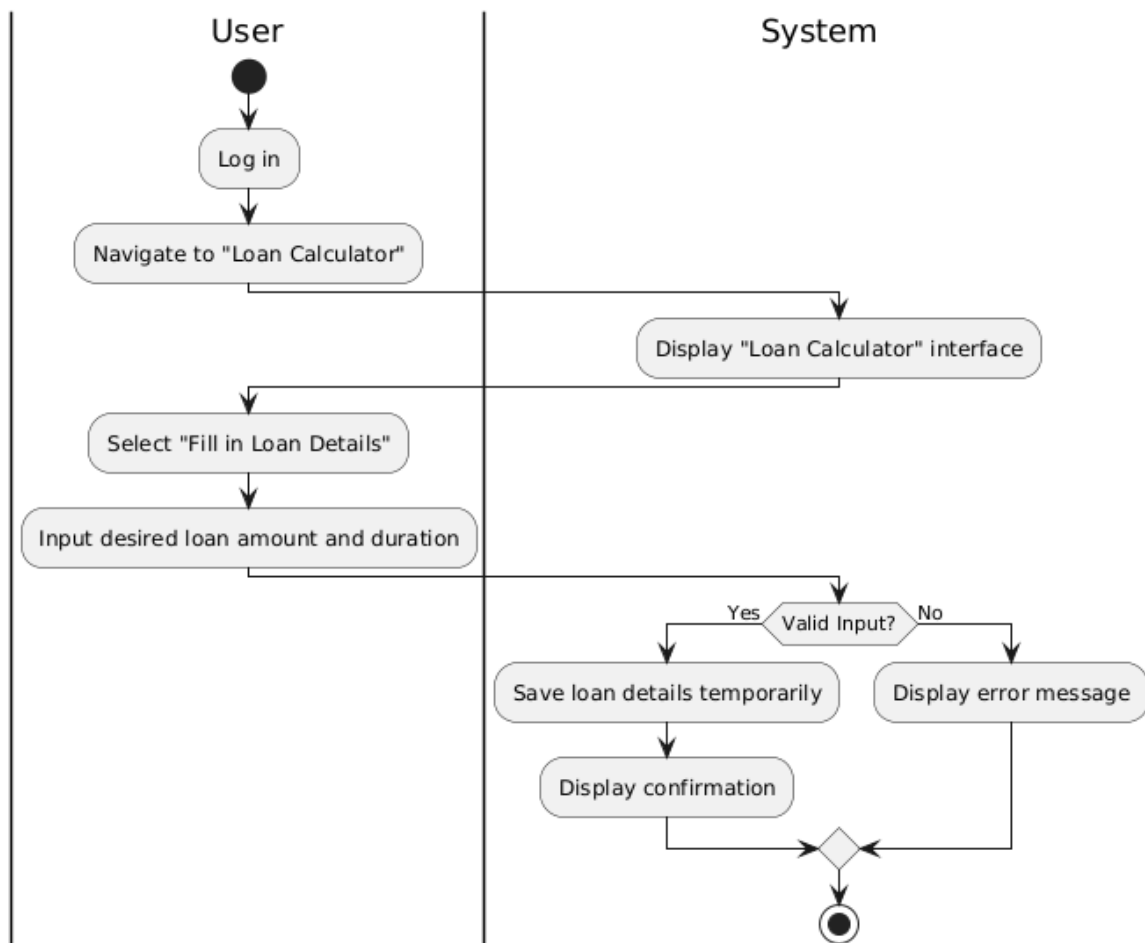


Figure x.x: Activity Diagram of US016

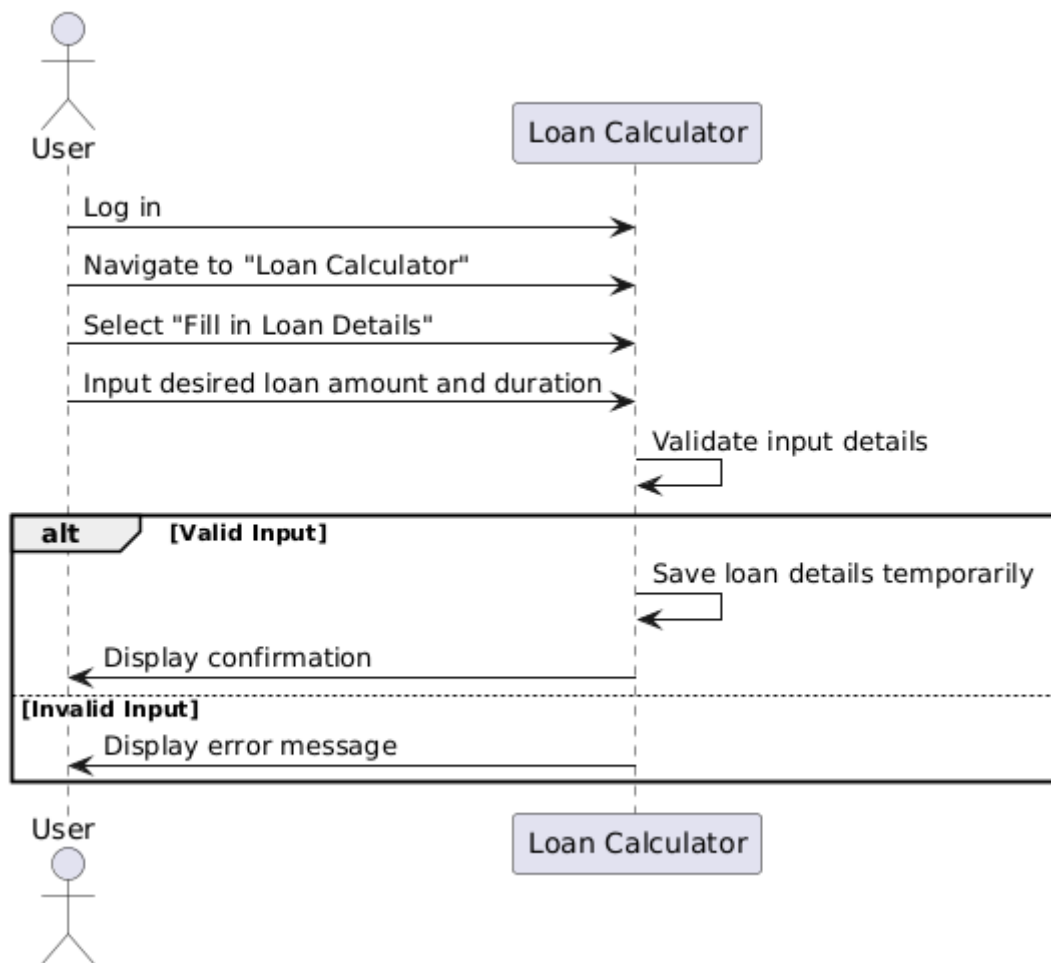


Figure x.x: Sequence Diagram of US016

2.4.15 US017: User Story <User Story 17>

Table 2.1: User Story Description for Generate Loan Repayment Plan

User story: Generate Loan Repayment Plan				
ID: US017				
User Story Description				
As	a	system	user,	
I	want	to	generate	a detailed repayment plan,
So that I can understand my financial obligations.				

Flow of events:

1. User fills in loan details (UC016).
2. User selects "Generate Repayment Plan."
3. System calculates monthly installments, total interest, and total repayment amount.
4. System displays the repayment plan, including a breakdown of each installment.
5. User can download or print the repayment plan.

Alternative flow:

1. If calculations fail due to invalid inputs, the system prompts the user to re-enter details.
2. If the user cancels the process, no repayment plan is generated.

Acceptance Criteria

Precondition: A repayment plan is displayed or downloaded for the user.

Postcondition: Valid loan details must be provided in UC016.

Exception flow:

1. Calculation error: System notifies the user to retry with correct inputs.
2. Server error: System notifies the user to try again later.

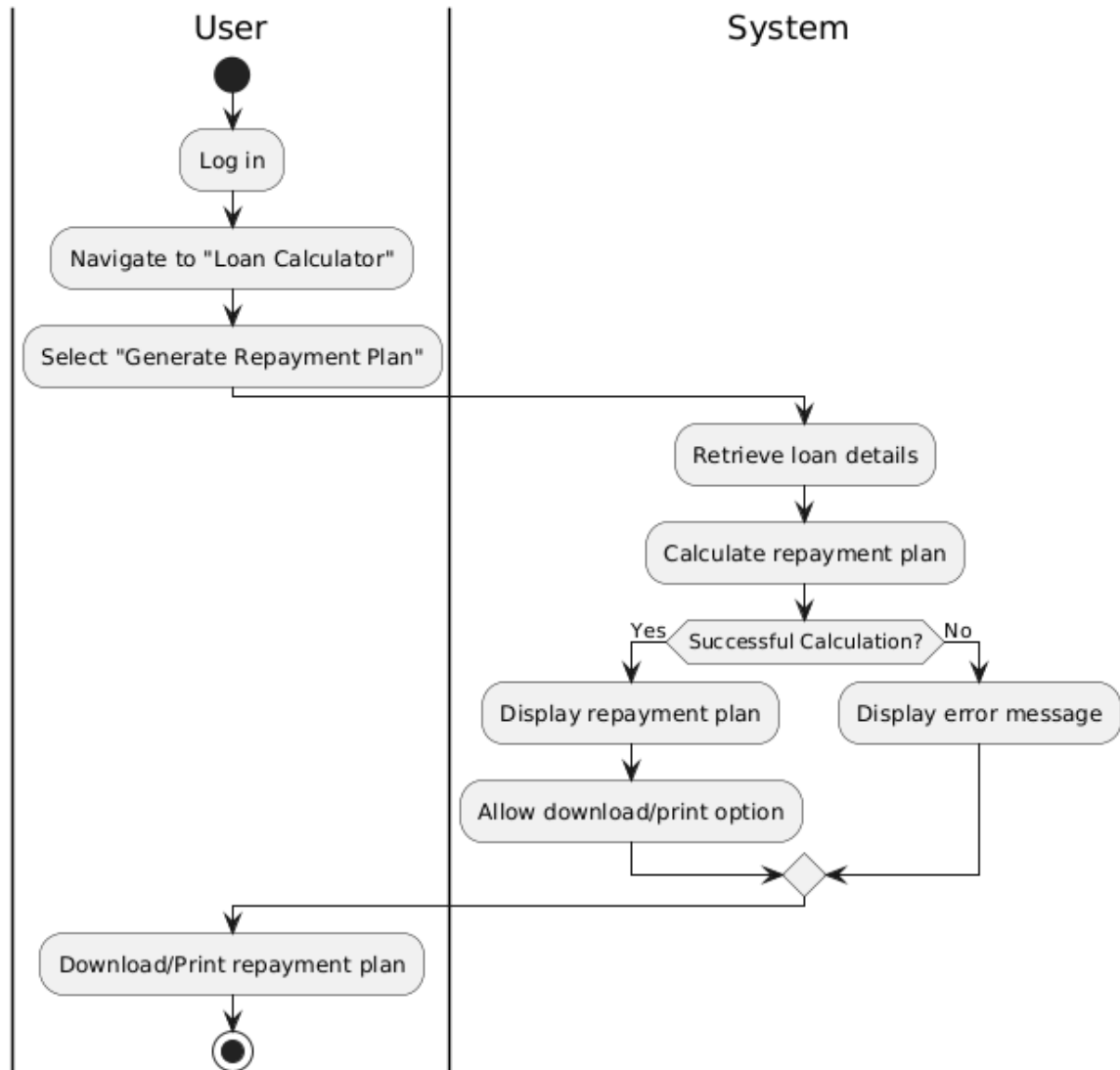


Figure x.x: Activity Diagram of US017

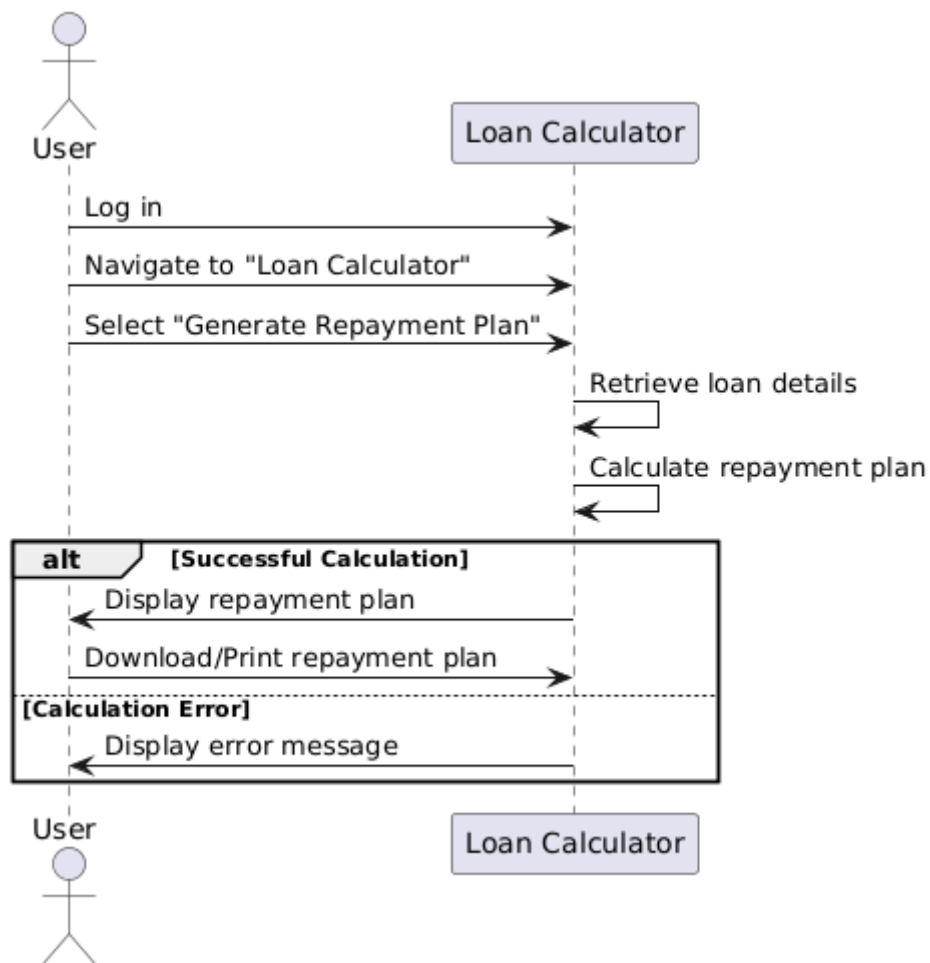


Figure x.x: Sequence Diagram of US017

2.4.15 US018: User Story <User Story 18>

Table 2.1: User Story Description for Loan Repayment

User story: Loan Repayment
ID: US018
User Story Description
As a member with an active loan, I want to view and make repayments for my loan, So that I can fulfill my financial obligations.
Flow of events:

1. User logs into the system.
2. User navigates to the "Loan Repayment" section.
3. System displays loan details, including outstanding balance and due dates.
4. User selects an installment to repay.
5. System processes the payment and updates the repayment status.
6. User receives a confirmation receipt.

Alternative flow:

1. If no active loans exist, the system displays a message: "No active loans found."
2. If the payment fails, the system notifies the user and prompts them to retry.

Acceptance Criteria

Precondition: The user must have an active loan with pending repayments.

Postcondition: The loan repayment status is updated, and the user receives confirmation.

Exception flow:

1. Payment gateway error: System notifies the user and suggests retrying.
2. Insufficient funds: System notifies the user and cancels the payment.

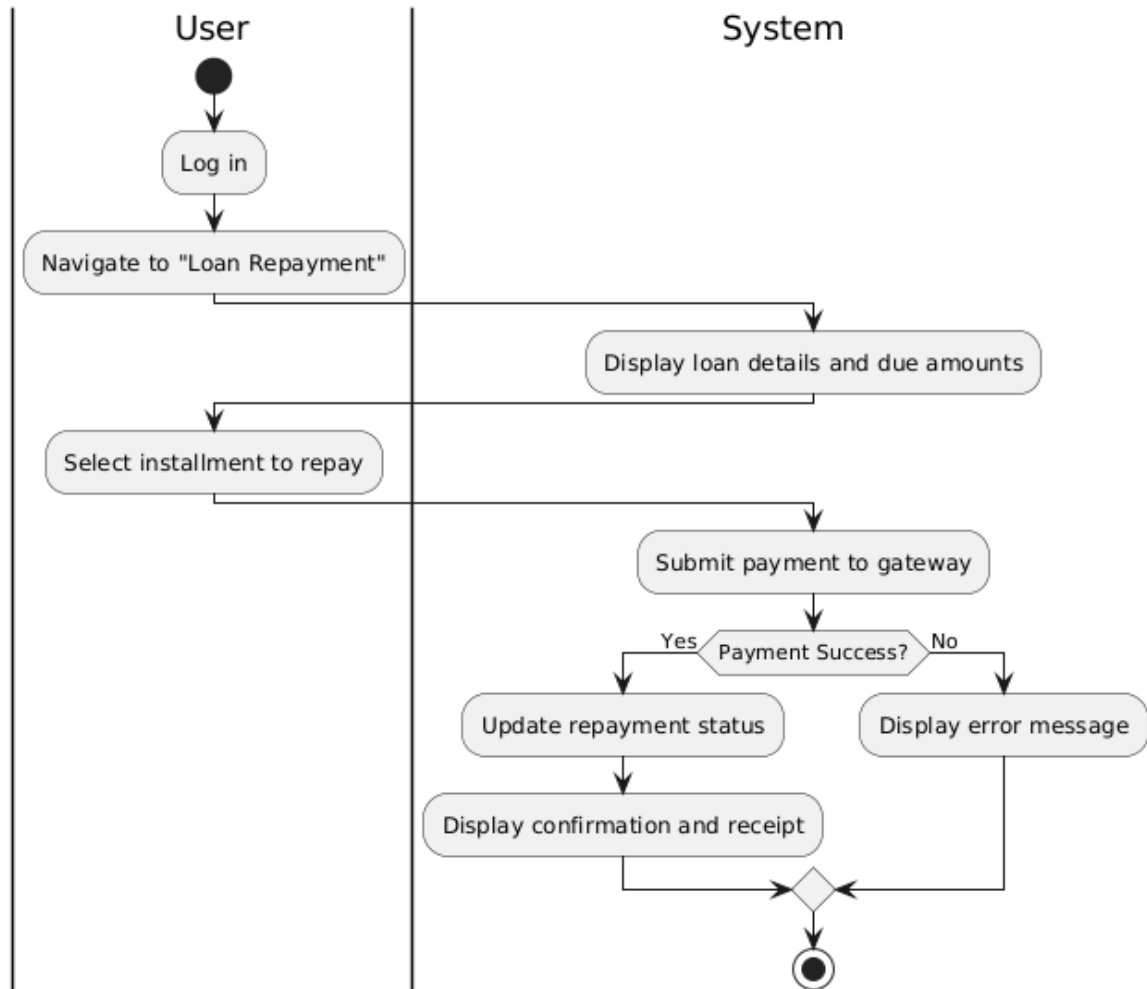


Figure x.x: Activity Diagram of US018

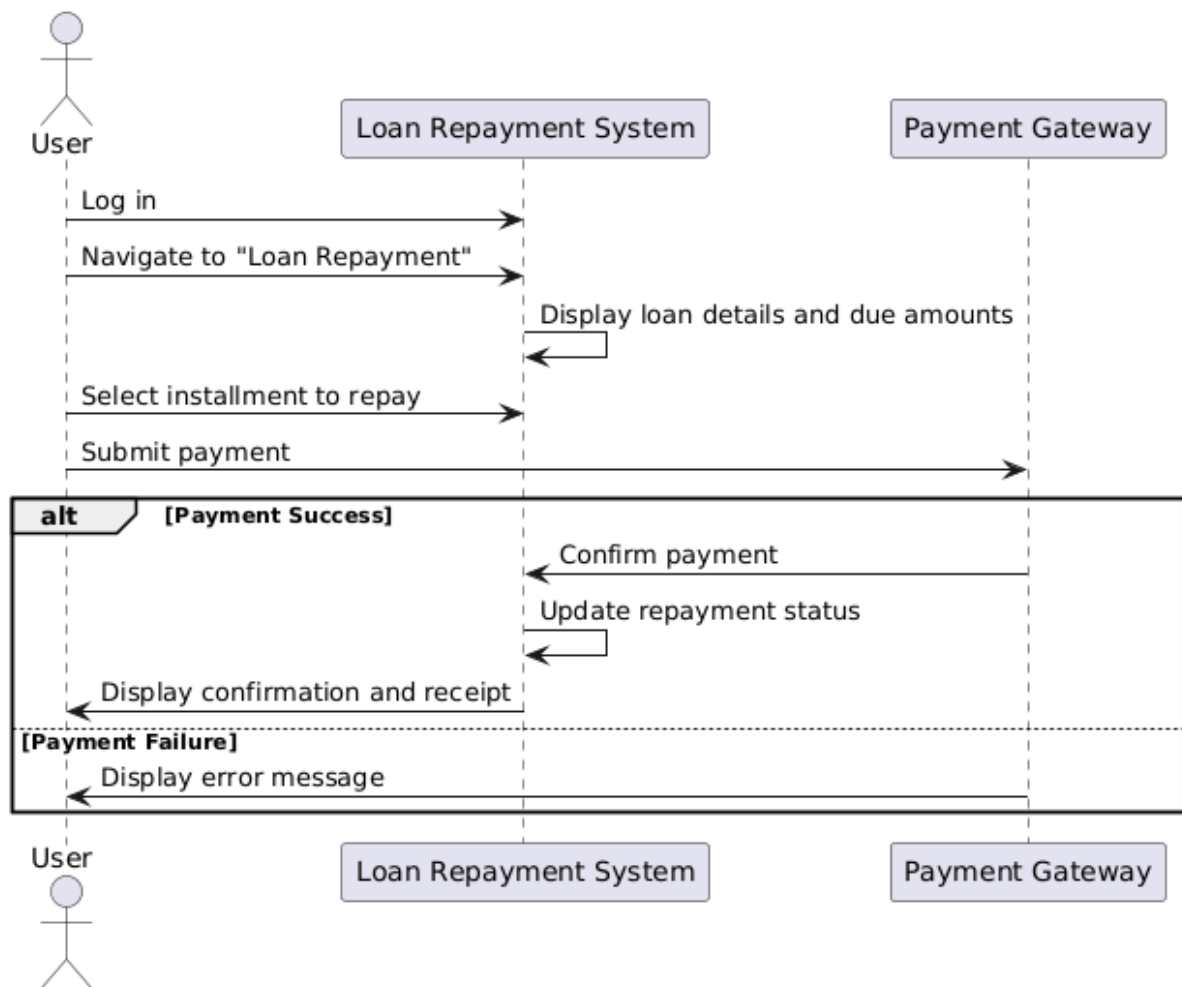


Figure x.x: Sequence Diagram of US018

2.4.15 US019: User Story <User Story 19>

Table 2.1: User Story Description for Pay Monthly Advisable Amount

User story: Pay Monthly Advisable Amount
ID: US019
User Story Description As a loan borrower, I want to pay the monthly advisable installment amount, So that I can keep my repayment schedule on track.
Flow of events:

1. User logs into the system.
2. User navigates to the "Loan Repayment" section.
3. System displays the advisable monthly amount based on the repayment plan.
4. User selects the payment option for the advisable amount.
5. System processes the payment and updates the repayment status.
6. User receives a confirmation receipt for the payment.

Alternative flow:

1. User inputs the desired payment amount.
2. System recalculates the repayment schedule.

Acceptance Criteria

Precondition: The user must have an active loan with an advisable installment due.

Postcondition: Member need to wait for their application results before creating a new one.

Exception flow:

1. Payment gateway error: System notifies the user to retry.
2. Insufficient funds: System notifies the user and cancels the payment.

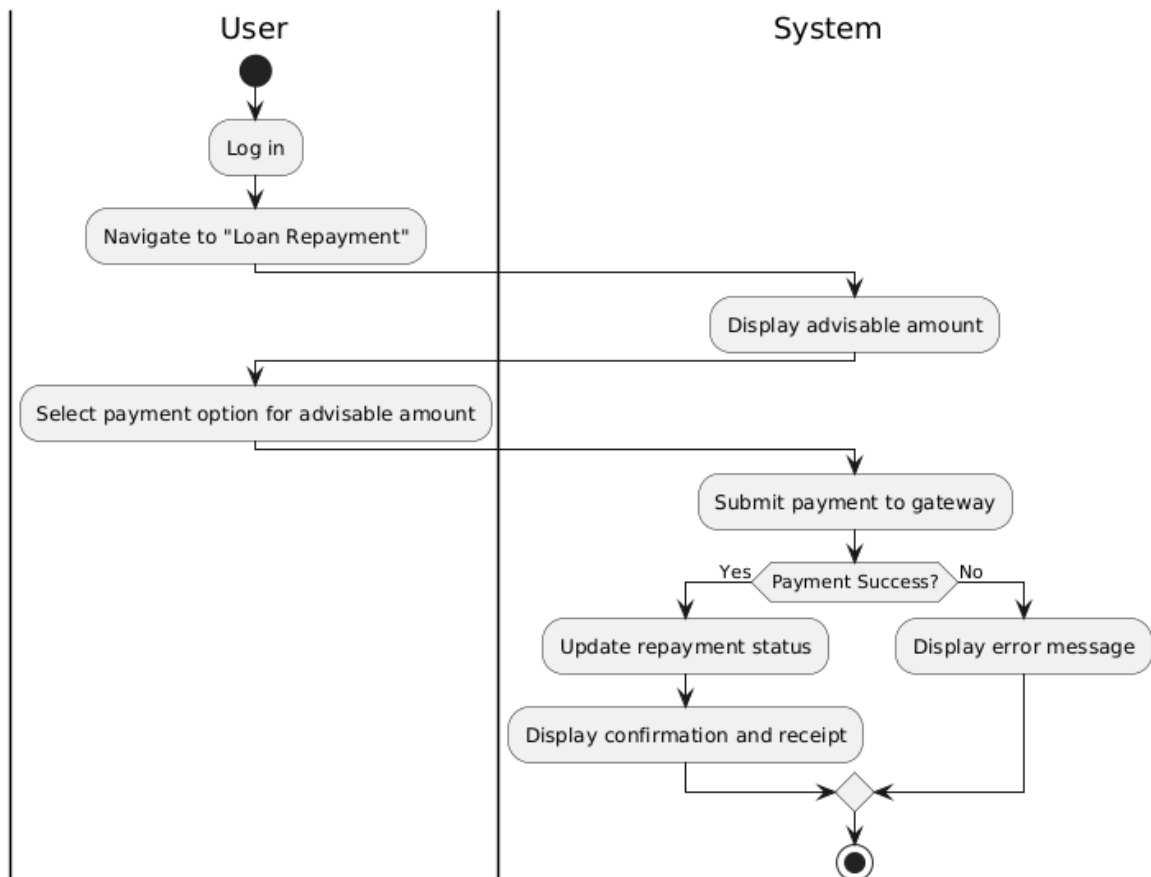


Figure x.x: Activity Diagram of US019

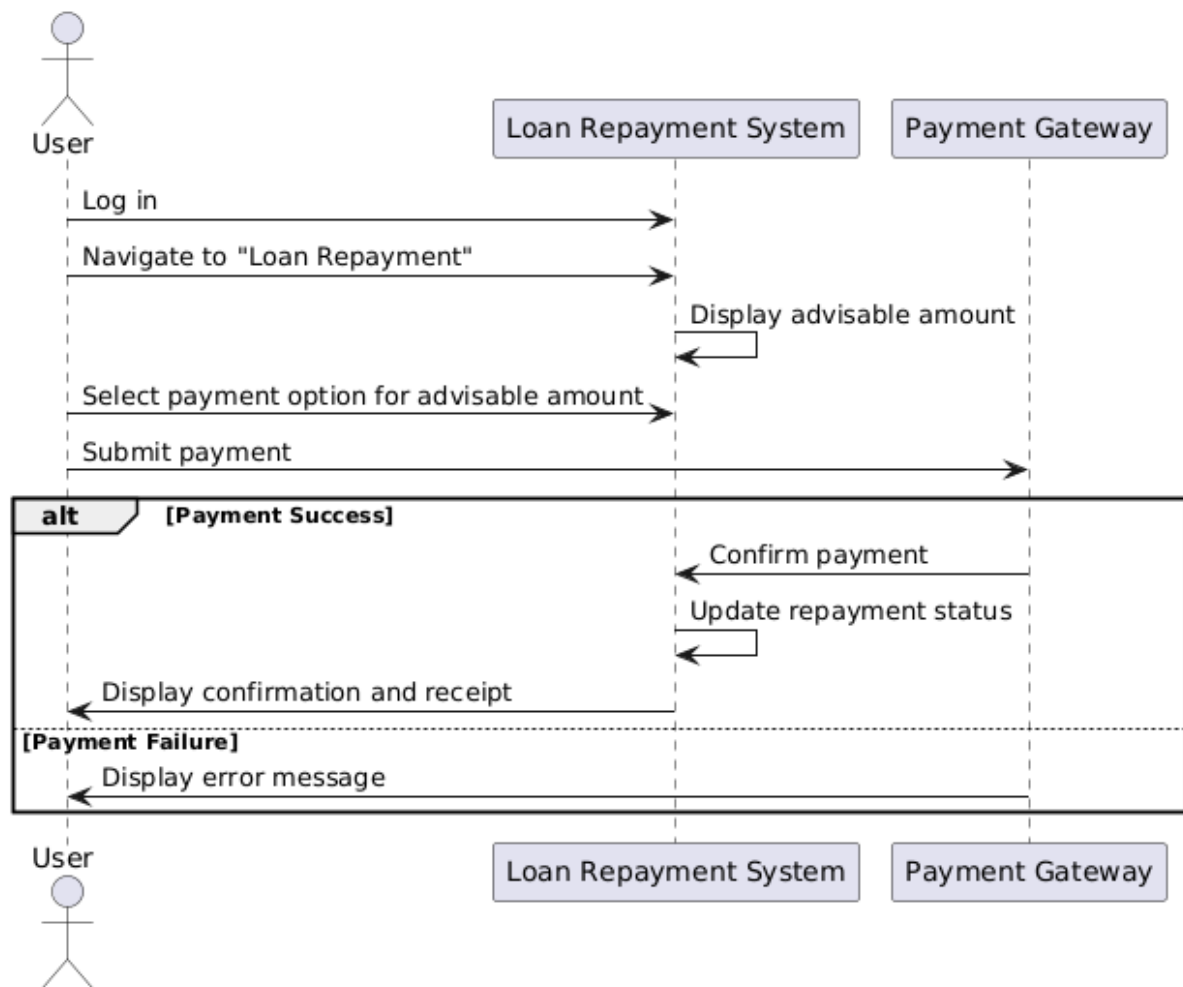


Figure x.x: Sequence Diagram of US019

2.4.2 US017: User Story <Generate Loan Repayment Plan>

2.4.3 US018: User Story <Loan Repayment>

2.4.4 US019: User Story <Pay Monthly Advisable Amount>

X.X.16 US016: User Story <Fill in Loan Details>

X.X.16 US017: User Story <Generate Loan Repayment Plan>

X.X.16 US018: User Story <Loan Repayment>

X.X.16 US019: User Story <Pay Monthly Advisable Amount>

2.5 Performance and Other Requirements

The purpose of this section is to define how the system should behave and operate to meet KADA's user and organizational needs effectively. These requirements focus on system speed, reliability, usability, and security, ensuring a seamless experience of applicants, members, admins, and board members (ALK). By addressing these factors, the system ensured its long-term sustainability and growth.

Performance Requirements

To meet the expectations of KADA's users, the system must deliver consistent and efficient performance:

1. Response Time:

- 95% of user interactions, such as submitting forms or checking application status, should be completed within 2 seconds.
- Administrative actions, like searching applicant details, must not exceed 5 seconds, even during peak usage.

2. Throughput:

- The system must handle up to 500 simultaneous users at any given time, reflecting KADA's current operational limits.
- Scalability should allow for future expansion to support 1,000 concurrent users at the cooperative grows.

3. Availability:

- The system must maintain an uptime of 99.9%, equating to less than 8 hours of downtime annually.
- Important services like payment processing and application submissions must remain operational during recovery periods.

4. Data Backup and Recovery:

- Automatic data backup should occur nightly at 2:00 AM.
- The system should recover from problems within 1 hour.

System Qualities

The system must be easy to use, reliable, and secure for all users.

1. Usability:

The interface should be simple and user-friendly. It should also support multiple languages, like English and the local language like Malay, Chinese and Tamil.

2. Reliability:

The system should keep all data safe, even if there is a crash or power outage.

3. Security:

- Sensitive information must be protected using strong encryption methods.
- Password should be strong (at least 12 characters, including letters, numbers, and symbols).
- Admins and board members (ALK) should use extra security like two-factor authentication.

2.6 Design Constraints

This section lists the limits and rules the system must follow. These include the environment it works in, the hardware it needs, and the security measures it must have. Following these rules ensures the system runs smoothly and fits with KADA's needs.

Environmental Constraints

The system should work well on different devices and in standard office conditions.

- **Devices:** The system must work on desktops, tablets, smartphones, and laptops. It should also support Android and iOS.
- **Environment:** It should work in temperatures between 10°C and 35°C and humidity between 20% and 80%.

Hardware Constraints

The system must work with KADA's current hardware and be accessible to users with basic devices.

- **Servers:**
 - Must have at least 8GB RAM, dual core processors and 500GB storage.
 - Backup servers should be used to avoid downtime.
- **User Devices:**
 - Should work on devices with 2GB RAM and modern browsers like Chrome, Firefox and Safari.
 - Must support printers and scanners for document processing.

Security Constraints

The system must protect sensitive user data and ensure safe transactions.

- All communication between users and the system must be secured using HTTPS.
- Payment must follow safe standards like PCI DSS (Payment Card Industry Data Security Standard)
- Only authorized users should have access to important data. Logs of user actions must be kept for at least 6 months.

Compatibility Constraints

The system must fit well with existing tools and ensures smooth integration.

- **Data migration:** All old data must be transferred without any loss.
- **Third Party Tools:** It must work with payment system like Paypal and local banks.
- **Browsers:** The platform should work on all major browsers and adjust to different screen sizes.

Performance Constraints

The system must operate within certain limits while remaining reliable

- **Users:** The system can handle up to 500 users at the same time.
- **Speed:** Search results should load within 3 to 5 seconds, even during busy time.
- **Backup:** Data should be backed up nightly, with the ability to restore within 1 hour.

Regulatory Constraints

The system must follow all laws and regulations to protect users and stay compliant.

- **Data privacy:** It must follow data protection laws like GDPR (General Data Protection Regulation) if applicable.
- **Financial Regulations:** Loan and financial features must comply with local rules.